Corvinus University of Budapest

Forecasting the 2026 Hungarian Parliamentary Election

Integrating Poll Aggregation, Structural Fundamentals, Online Media Consumption and Seat Projection

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Abstract

The 2026 Hungarian parliamentary election is an unusually demanding forecasting problem. Hungary's mixed-member system allocates 106 single-member districts and 93 national list seats under a compensation vote mechanism that amplifies small national differences into large seat swings. The domestic polling landscape is limited in volume and raises questions about systematic pollster bias, while the current electoral boundaries have been in place only since 2014, leaving at most four comparable election cycles to learn from. This study asks whether the outcome of the 2026 election can be forecast from available data, how sensitive the seat distribution is to small changes in national vote share and how large the irreducible uncertainty around any prediction is.

The forecasting framework consists of five linked layers. A national poll model aggregates available surveys through a simple sample-weighted average and a house-effect-corrected Kalman smoother. A national fundamentals prior (baseline estimate) is constructed bottom-up from district-level Ridge regression on demographic predictors and serves as a structural anchor against short-term polling swings. A district lean model with county-level random intercepts translates the national forecast to all 106 districts, extended by a recent correction layer that incorporates lagged election results and pro-government digital media reach. A candidate-gap model estimates the difference between candidate and list vote shares using hierarchical mixed-effects regression. A Monte Carlo simulation with 1,500 draws propagates uncertainty through the full pipeline under the Hungarian compensation vote rules and D'Hondt list seat allocation. All vote shares are treated as compositional data using additive log-ratio transformations and the framework is validated through Leave-One-Election-Out backtesting.

Before the election, the combined forecasts placed the two main blocks within two percentage points of each other, with a government majority probability between 39 and 43 percent. The April 12 result delivered a decisive TISZA victory at 53 percent of the list vote, producing a supermajority of 138 seats. The structural prior was the primary error source, pulling government support upward against the actual direction of the result.

Keywords: Election Forecasting, Poll Aggregation, Parliamentary Elections, Monte Carlo Simulation, Hungary

Contents

I. Introduction

In many countries, including the United States, election forecasts are now a regular feature of public debate and media coverage (Gelman, 2024). In Hungary, there are far fewer transparent, data-driven election forecasts, partly because it is a small country that attracts limited attention from global audience, even though parliamentary elections are high-stakes events that shape the country's political and economic future.

The 2026 Hungarian parliamentary election is especially interesting to study. The electoral system is complicated. Seats come from both single-member districts (OEVKs in Hungarian) and national party lists (Government of Hungary, 2025). Small changes in support in a few marginals can make a big difference to who forms the government (National Research Council, 2006, pp. 29–33). The political landscape is also unusual, with a long-standing governing party, a fragmented and changing opposition and strong regional differences between districts (Kóczián, 2025). Fidesz–KDNP has held a parliamentary supermajority since 2010 but now faces TISZA as a newly consolidated challenger for the first time, while Mi Hazánk sits to the right of the government and DK polls near the parliamentary threshold (see Figure A.1 in Appendix VIII.4.). On top of this, the information environment is not ideal, some pollsters are seen as politically aligned and the overall number of high-quality, independent surveys is limited (if such a category can be meaningfully identified at all). Beyond polling quality, the electoral history under the current mixed-member system is itself very short. The district boundaries and compensation vote rules have been in place only since 2014, leaving at most four comparable election cycles to learn from, which limits how much any data-driven model can extract from the past.

These features make Hungary a challenging but important test case for election prediction. A careful, data-based forecast can help answer simple but crucial questions: How close is the race likely to be? How sensitive is the final outcome to small changes in public support? At the same time, the exercise can reveal how much we can trust the available polling data and how large the uncertainty around any prediction really is.

The central research question of this study is whether the outcome of the 2026 Hungarian parliamentary election can be forecast from available data and how sensitive the seat distribution is to small changes in national vote share. The aim of this thesis is to build and evaluate a forecasting approach tailored to the Hungarian context, using past election results, available polls, basic economic indicators and media usage data. Beyond producing a forecast for 2026, the broader motivation is to provide a transparent, well-explained framework that can be used to think more clearly about Hungarian elections, their likely outcomes and the limits of what data can tell us in advance.

The central finding has two parts. Before the election, the combined forecast placed the government and the main opposition within two percentage points of each other at the national list vote level. At the seat level, the probability of a government parliamentary majority was between 39 and 43 percent, while the probability of a two-thirds supermajority ranged from 1 to 19 percent. The April 12 result resolved the uncertainty decisively. TISZA received 53 percent of the national list vote against 38 percent for the government, a gap of nearly 15 percentage points, which translated into 138 seats for the opposition and 55 for the government. The structural prior was the primary source of forecast error, pulling the combined estimate toward government support in a year when demographic and economic patterns from the previous election cycles did not carry forward to 2026.

The remainder of this paper is organised as follows. Section II reviews the election forecasting literature across the five streams that directly shaped the methodology. Section III describes the data sources, the analysis sample and the variable construction. Section IV presents the full methodological pipeline. Section V reports the results layer by layer, compares the two parallel poll paths and closes with an after-election accuracy assessment against the April 12 outcome. Section VI concludes.

II. Literature Review

The election-forecasting literature is very broad, but for the current work the most useful studies answer five practical questions. First, how should polls be aggregated and how large can poll error be. Second, how should polls be combined with slower structural information. Third, how can national information be translated to districts when local polling is weak. Fourth, how can vote forecasts be turned into seat forecasts in a multi-party system. Fifth, how digital media consumption data can serve as a meaningful local-level predictor of political support. The first four questions follow the standard architecture of modern election forecasting and all four are directly relevant here because the Hungarian parliamentary election is multi-party, mixed-member and geographically structured. The fifth question is the specific contribution of this study, because Hungary's media landscape is widely described as heavily government-aligned (ATLO Team, 2022b), which, if that description holds, could create a geographic media signal that is unusually clean and measurable by international standards.

II.1. Poll Aggregation And Poll Error

Election polls can share systematic errors because many polling institutes use similar fieldwork methods, similar respondent screens and similar pollster specific tendencies. As a result, even a large poll average may still reflect a common bias that simple aggregation cannot remove (Shirani-Mehr et al., 2018). Shirani-Mehr et al. (2018) quantify this problem across 4,221 polls from 608 US state-level elections, where the root mean square error of a single poll is roughly twice the reported margin of error and polls in the same election often share a common component of error, meaning all polls in the same race can miss the result in the same direction. The paper finds an average absolute election-level bias of about two percentage points.

These findings define two practical approaches. The simple approach takes a weighted average of recent polls, which is transparent and robust but leaves house effects uncorrected. The richer approach corrects each poll for estimated pollster bias and tracks latent support over time, as implemented by Linzer (2013), Heidemanns et al. (2020) and The Economist (2020). Heidemanns and the Economist framework explicitly adjust for systematic pollster bias, while Linzer (2013) mainly show a dynamic poll plus fundamentals model and note that averaging across pollsters can reduce house effects.

II.2. Polls And Fundamentals

Polls carry little information about the final result when they are conducted far from election day. Erikson and Wlezien (2008) show this directly for US presidential elections, where economic leading indicators predict the vote better early in the election year while party preference polls become increasingly informative only in the final weeks and Brown and Chappell (1999) confirm that combining both sources consistently reduces forecast error relative to either alone. Linzer (2013), Stoetzer et al. (2019), Heidemanns et al. (2020) and The Economist (2020) all implement this logic in a shared structure, where a fundamentals prior[1] sets the starting forecast before any campaign polls arrive and its weight shrinks as polls accumulate.

II.3. Subnational And Constituency Forecasting

In systems with single-member districts, the local distribution of support matters at least as much as the national total, yet local polling is rarely available at district level. The subnational forecasting literature addresses this by combining survey data with demographic and geographic structure, most commonly through multilevel regression and population weighting. Kiewiet de Jonge et al. (2018) show that state-level estimates produced this way can be highly accurate and Lauderdale et al. (2020) confirm the same for UK constituency forecasts, correctly indicating that the Conservatives would fail to secure a majority in 2017 (YouGov, 2017). The key finding across applications is that local precision gains come from combining geographic borrowing with constituency-level predictors. A simpler model without these sources of structure performs substantially worse (Hanretty et al., 2018) and Selb and Munzert (2011) confirm the value of minimal geographic information for German federal constituencies.

[1] The term *prior* is used here in the sense of a structural baseline estimate derived from demographic and economic predictors, independently of polling data.

For Hungary specifically, Kovacs and Vida (2015) show that spatial patterns in vote support are strong and persistent after the 2014 electoral reform. Government support is stronger in rural areas while the opposition holds urban strongholds, which is a direct empirical input to the district lean model in this study. Erfort et al. (2026) contribute a further relevant design choice. An explicit district-level candidate vote layer, directly analogous to the candidate layer used here.

II.4. Multi-Party Composition And Seats

In a multi-party system the vote shares of all parties must sum to one and remain non-negative. Aitchison (1986) establishes the statistical framework for this type of constrained data through log-ratio transformations that remove the constraints without losing information. Stoetzer et al. (2019) apply this to German federal elections and show that forecast errors across parties are correlated. A model that overestimates one party tends to underestimate the others, so uncertainty draws must be taken jointly across all parties rather than independently for each one.

Erfort et al. (2026) carry the multi-party logic further to the seat level for the German federal election of 2025. Their central finding is that national vote shares alone are a poor guide to the seat distribution, because seat allocation also depends on district winners, threshold rules and the possibility that some district winners do not receive a parliamentary seat. This same non-linearity is also a central challenge in the Hungarian context, where small changes in national vote shares can translate into much larger differences in parliamentary seat outcomes.

II.5. Media Effects And Digital Audience Data

This study also introduces a county-level digital media reach as a district-level predictor of political support. This choice is motivated by three main arguments.

The first establishes that media exposure genuinely changes voter behaviour. Gerber et al. (2011) conduct the first large-scale randomised field experiment on political advertising, deploying approximately 2 million US dollars of television and radio advertising experimentally across media markets during a 2006 US gubernatorial campaign. The exogenous assignment shows that televised campaign advertising has a large effect on voter preferences, but it decays almost completely within two weeks of the final ad. Martin and Yurukoglu (2017) provide causal evidence for a related channel, namely the political slant of media content and not just its volume. Using cable channel positions as an instrument for Fox News viewership, they find that politically slanted media leaves a detectable imprint on election results and contributes to political polarization.

The second establishes that digital audience data can serve as a usable input with appropriate normalisation. Wang et al. (2014) show this through an Xbox user survey that was heavily skewed toward young males. After adjustment for population composition, the estimates are competitive with professional polling averages. This suggests that even large digital data sources can contain useful electoral information, despite not being fully representative of the electorate.

The third focuses on Hungary, where the findings discussed above are especially relevant because political support has a spatial structure and media reach can therefore capture meaningful geographic differences. Kovacs and Vida (2015) show that Hungarian party support has a strong spatial structure, with a sharper urban rural divide after the electoral reform. Government support is stronger in many rural constituencies, while the opposition holds stronger positions in Budapest and some larger cities.

III. Data

In this section I describe the data sources, the structure of the analysis sample and the variables that enter the forecast. First, I describe each source, then explain how the observations are organized and finally I define the variables used in the models.

III.1. Data Sources

I rely on six main data sources, which will be described below.

III.1.1. Polling Data

The polling data comes from the Vox Populi election guide maintained by Tóka Gábor (Tóka, 2019b). This database collects Hungarian pre-election surveys from all publicly available polling institutes. The current version available contains 1,727 polls recorded between October 2017 and April 2026 (the database also includes 16 polls from 2014, but this small number is not significant for the analysis). Each poll record includes the pollsters' name, the fieldwork start and end dates, the geographic focus, the interview mode, the sample size, the respondent base and the reported vote shares for all listed parties.

The interview modes in the database are online, telephone, hybrid, IVR, personal interview and mobile application surveys. The two main respondent bases are "total population" and "sure voters". The total population base includes all adults regardless of their voting intention, providing a broad view of general political sentiment. In contrast, the "sure voters" base consists only of respondents who express a high likelihood of participating in the election.

The geographic focus distinguishes between national representative surveys, European Parliament specific polls, local district or city level studies and hypothetical electoral scenarios. Sample sizes range from a few hundred to 60,000, with a median of 1,000 respondents. The 34 polling institutes include Medián, Századvég, Publicus, Nézőpont, Republikon, IDEA, Ipsos and several smaller firms. The exact list can be found in Appendix VIII.1..

III.1.2. Election Results

Official election results at the settlement, electoral district and national levels come from two sources.

At the settlement and electoral district levels, I use the Vox Populi results database compiled by Tóka Gábor (Tóka, 2018, 2019a, 2022, 2024). These are based on the official national results but they are compiled more nicely and are easier to access and work with. One notable limitation is that these databases only contain votes cast at domestic polling stations and do not include the letter votes from non-resident citizens. On one hand, this is problematic because a fraction of votes are missing. On the other hand, the mail voters do not affect the individual district results that truly decide the election according to National Election Office of Hungary (2026b) and Kovalcsik (2025). Furthermore, this exclusion ensures model consistency. Polling data or socio-economic fundamentals are not available for the non-resident population and including them would only add noise to the district-level estimates. By focusing on the domestic results, the target variables and the predictors remain aligned within the same geographic and demographic framework.

These datas further contain party-level vote counts, valid votes, electorate sizes and turnout figures for each settlement and single-member district. I use settlement-level data as the finest available unit because it allows to re-aggregate historical results to the new 2026 district boundaries, as explained in the next subsection.

At the national level, the seat totals and aggregate vote shares for the European Parliament elections were transcribed from the official results published by the National Election Office (National Election Office of Hungary, 2019, 2024). The parliamentary seat distributions were compiled from the official results of the National Election Office and the detailed summaries provided by Wikipedia (National Election Office of Hungary, 2022; Wikipedia, 2018).

III.1.3. Settlement-Level Statistics

The socio-economic and demographic data at settlement level comes from the Hungarian Central Statistical Office (KSH) interactive database (Hungarian Central Statistical Office, 2026). Since the available data for 2026 or even 2025 is very limited, each election year is matched with the settlement-level statistics published two years prior to the vote. For example, the 2018 election uses 2016 data, the 2019 election uses 2017 data and so on. Furthermore, this two-year lag prevents the model from incorporating economic signals that might already be influenced by the election campaign. These variables therefore function as a purely structural prior that describes the long-term demographic and infrastructure conditions of the electoral districts.

The raw settlement-level variables include total population, population aged 0–14, population aged 65 and above, permanent residents by age and gender, registered unemployed by duration and education, budget revenue and expenditure, number of dwellings built, general practitioners, paediatric doctors, registered cars, internet subscriptions by technology type and several other indicators which can be found in Appendix VIII.2., along with their descriptive statistics.

III.1.4. Territorial Labour Income

County-level average worker income comes from a separate KSH source (Hungarian Central Statistical Office, 2026). The original data is in nominal million HUF. To make the values comparable across time, I deflate them to real 2010 prices using the consumer price index from the World Bank (World Bank, 2025). The time alignment follows the same logic as the settlement statistics. The income data is matched to each election, currently with a three year lag, because of data availability (for example, the 2015 data is matched to the 2018 election). Further description in Appendix VIII.2..

III.1.5. Media Consumption Data

Online media consumption data comes from the DKT – Gemius audience measurement system, who were so kind to give access to their database. Since the data is only available for the last 5 years I could only use for 3 elections. One for the 2022 parliamentary election (DKT – Gemius, 2022), one for 2024 (DKT – Gemius, 2024) and one for 2026 (DKT – Gemius, 2026). I gathered a county- and place-size-level data for many online outlets. The target audience is the domestic population aged 16–75 with no additional consumer filter applied.

The tracked 27 outlets are covering a broad spectrum of the Hungarian digital landscape. These include social media platforms like Facebook and TikTok, alongside major news portals such as 24.hu, Index, Origo, Telex and HVG. The data also incorporates specialized and broadcast-related sites including Hirado, RTL, Mandiner and Nepszava.[2] Each outlet is described by three consumption metrics: real users, page views and average time per view. Real users measure how many distinct individuals visited a given outlet during the selected period, page views capture how many times its content was opened and average time per view indicates how much time was spent (in seconds) on a single content view on average. Together, these measures distinguish audience size, consumption intensity and the average depth of attention received by the outlet.

[2] In this paragraph, I followed the stylistic and grammatical guidance provided by generative AI.

These consumption data are downloaded in three time windows: first one covers the period from 12 to 7 months before the election, second one from 6 to 3 months before and the third one represents the final two months leading up to the vote. Further details on data availability for each outlet are provided in Appendix VIII.3..

Table 1 provides a clear summary of the data sources.

Source	**Content**	**Role in the forecast**
Vox Populi poll database	Pre-election survey results	National poll model
Vox Populi election results	Settlement and district-level results	Target variable for all layers
KSH	Settlement level statistics	Structural predictors
KSH	County-level income	Economic predictor
World Bank	Consumer price index for Hungary	Deflation of nominal income to real values
DKT–Gemius	Online media consumption by county and place size	Dynamic media signal in district lean model

Table 1: Overview of data sources

Source: Compiled by the author.

III.2. Analysis Sample

III.2.1. Unit of Analysis and Aggregation

The unit of analysis is the single-member electoral district (OEVK).[3] Hungary has 106 such districts for parliamentary elections (National Election Office of Hungary, 2026b). Because the finest available data is at the settlement level, all settlement-level statistics and election results are first prepared at the settlement level and then aggregated up to the OEVK level using voter population weights from the National Election Office files (National Election Office of Hungary, 2026c).

This settlement-first approach is necessary because the OEVK boundaries changed for the 2026 election (National Election Office of Hungary, 2026c; Weiler, 2024). By working at the settlement level and then re-assigning settlements to the new 2026 districts based on official correspondence data (National Election Office of Hungary, 2026c) all historical data can be placed on a consistent geographic basis. The voter population weights ensure that larger settlements within a district receive proportionally more influence in the aggregated OEVK-level variables. Without this step, the historical district data would not be comparable to the

[3]The only exception is the national poll model, where the observations are individual polls.

2026 forecast districts. Historical election results are therefore re-expressed too on the 2026 district map for district-level analysis and forecasting.

III.2.2. Elections Covered

I cover the 2018, 2022 parliamentary elections and the 2019, 2024 European Parliament elections as well. I selected these cycles because both polling data and statistical variables are consistently available. I excluded the 2014 elections not just because the missing data, but I think those results are now too distant to provide a relevant signal for today's political landscape.

III.2.3. Political Blocks

To ensure a consistent comparison across election cycles, where coalitions and alliances changed substantially, I group all parties into four political blocks to provide a more stable basis for the final forecast:

1. *Government:* Fidesz–KDNP in all elections.
2. *Opposition:* The main opposition parties, which are listed in table 2 by election year. In 2026 only TISZA is included, because DK–MSZP–P has largely lost its political relevance and — together with MKKP — is polling below the 5% parliamentary threshold under the Hungarian electoral system (Index.hu, 2026a) (see Figure A.1 in Appendix VIII.4.).
3. *"Radical" opposition:* Jobbik in 2018 and Mi Hazánk from 2019 onward. This block represents the right wing voters who are separate from both the government and the main opposition.
4. *Other:* All remaining parties and independent candidates.

This grouping is necessary because modelling individual parties across elections would be impossible when parties merge, split or disappear entirely between cycles (Stoetzer et al., 2019).

Year	Included Parties
2018	DK, MSZP–Párbeszéd, LMP, Momentum
2019	DK, MSZP–Párbeszéd, LMP, Momentum, Jobbik
2022	DK–MSZP–Párbeszéd–LMP–Momentum–Jobbik
2024	TISZA, DK–MSZP–P
2026	TISZA

Table 2: Composition of the opposition block by election year

Note: LMP is categorized within the "Other" group starting from the 2024 election because of its declining political relevance. Source: Compiled by the author based on own grouping rationale and data from ATLO Team (2022a).

III.3. Variables

III.3.1. Polling Variables

To ensure data consistency and relevance, the analysis sample is restricted to polls conducted within a 365-day campaign window prior to each election (Linzer, 2013; Stoetzer et al., 2019). Any records that are missing essential variables, like sample size, interview mode and respondent base, are excluded from the analysis. I included both the "total population" and "sure voters" respondent bases to capture the difference between general party popularity and those who are likely to vote. For example, the "sure voters" base often shows higher support for major parties because it excludes undecided people who are still part of the "total population" base.

I also only included national representative surveys and European Parliament specific polls, while excluding district-level (OEVK) or city-specific studies, because the number of local polls is too small to build a reliable separate model for every district and their inclusion in the national series would introduce local bias into the aggregate. These are important controls because different survey methods and geographic scopes can produce systematically different estimates for the same underlying support (Shirani-Mehr et al., 2018; Wang et al., 2014).

After these filters, the number of individual polls varied substantially across cycles: 122 for 2018, 24 for 2019, 93 for 2022 and 39 for 2024. For 2026 the sample contained 220 polls from 17 pollsters, covering the period from 18 April 2025 to 10 April 2026. A detailed description of the pollsters in the analysis sample can be seen in Table A.5 in Appendix VIII.5..

III.3.2. Election Result Variables

In parliamentary elections, I calculate both list and candidate vote shares. I preserve both types because list votes reflect general party support while candidate votes capture local candidate effects (Erfort et al., 2026). In European Parliament elections, only the list share is calculated. The methodology later explains how the gap between these two shares is modeled separately for parliamentary elections.

III.3.3. Structural Variables

From the raw settlement-level data, I aggregated back to OEVK level with a set of per-capita and share-based features. In case of the county level labour income variable, the county-level value is directly assigned to all districts within that county. During the filtering process, I only selected variables that were available for almost every district and every election year. The highly correlated features were also filtered out. The final predictor set includes demographic, economic, modernization and public service variables. The exact variables and their descriptive statistics can be seen in Table 3, where:

- *Log population* is the natural logarithm of total population and captures district size on a log scale.
- *Age 0–14 share* is the share of the population aged 0–14 in total population.
- *Age 65+ share* is the share of the population aged 65 and above in total population.
- *Unemployed people per 1,000 residents* is total registered unemployment per 1,000 residents.
- *University-educated unemployed per 1,000 residents* is registered unemployment among university graduates per 1,000 residents.
- *Budget balance per 1,000 residents* is budget revenue minus budget expenditure, expressed per 1,000 residents. Positive values indicate a fiscal surplus.
- *Cars per 1,000 residents* is the number of registered passenger cars per 1,000 residents.
- *New car registrations per 1,000 residents* is the number of newly registered cars per 1,000 residents and serves as a proxy for recent economic activity.
- *Doctors per 1,000 residents* is the number of doctors per 1,000 residents.
- *Private dwellings built per 1,000 residents* is the number of privately built dwellings per 1,000 residents.

Feature	2018 NA	2019 NA	2022 NA	2024 NA	2026 NA	Mean	Median	SD	Min	Max
Age 0–14 share	100.0%	100.0%	100.0%	100.0%	100.0%	0.145	0.143	0.016	0.089	0.188
Age 65+ share	100.0%	100.0%	100.0%	100.0%	100.0%	0.199	0.198	0.023	0.143	0.258
Budget balance per 1,000 residents	100.0%	100.0%	100.0%	100.0%	100.0%	5,063.587	2,933.861	65,991.261	-413,058.207	534,617.354
Cars per 1,000 residents	100.0%	100.0%	100.0%	100.0%	100.0%	395.535	394.152	64.516	224.710	676.379
Doctors per 1,000 residents	100.0%	100.0%	100.0%	100.0%	100.0%	0.609	0.605	0.086	0.387	0.851
Log population	100.0%	100.0%	100.0%	100.0%	100.0%	11.416	11.417	0.114	11.107	11.683
New car registrations per 1,000 residents	100.0%	100.0%	100.0%	100.0%	100.0%	31.351	29.466	12.636	12.117	105.306
Private dwellings built per 1,000 residents	100.0%	100.0%	100.0%	99.1%	99.1%	0.756	0.468	0.885	0.000	6.762
Unemployed people per 1,000 residents	100.0%	100.0%	100.0%	100.0%	100.0%	26.924	22.318	16.866	5.266	91.827
University-educated unemployed per 1,000 residents	100.0%	100.0%	100.0%	100.0%	100.0%	0.762	0.608	0.524	0.125	2.882

Table 3: Non-missing share and pooled descriptive statistics of the selected structural predictors

Source: Compiled by the author.

It can be seen that the features appear broadly plausible overall and do not suggest major data quality problems.[4] I selected these variables not only because they are available, but also because they capture important aspects of the socio-economic and demographic landscape that can influence voting behavior. For example, the age structure (young and old share) reflects generational differences in political preferences, while the unemployment variables could capture economic distress that might affect support for the government or opposition. The budget balance indicates local fiscal health and public service quality, which are often linked to

[4]In this paragraph, I followed the stylistic and grammatical guidance provided by generative AI.

incumbent support (Erikson & Wlezien, 2008; Stoetzer et al., 2019). Car ownership serves as a proxy for wealth and modernization, while the doctor and dwelling variables reflect access to public services and local development.

Three additional variables were considered in the initial shortlist but removed after the first correlation check. These were the general practitioners per capita (redundant with the total number of doctors), blue-collar unemployed per capita variable (highly collinear with total unemployment) and post offices per capita (strongly correlated with population and district size).

The correlation matrix of the final variables is shown in Appendix VIII.6.. While some correlations are stronger than typical, there is no harmful multicollinearity. The stepwise "staircase" model selection process — detailed later in the district fundamentals section of the methodology (Section IV.2.) — addresses these relationships by grouping variables into logically distinct layers.

III.3.4. Media Variables

First, the media data was matched to the electoral districts (OEVK) based on county and settlement size categories. I include a wide set of online media outlets because media exposure can affect voter preferences in ways that structural demographic variables alone would miss (Gerber et al., 2011; Martin & Yurukoglu, 2017). From the Gemius audience reports, I construct media features as per capita log transformed values:

$$\text{media\_feature} = \log\left(1 + \frac{\text{raw metric}}{\text{pop\_total}}\right), \tag{1}$$

where *media_feature* is the final predictor used in the model, *raw metric* refers to the Gemius audience count (such as real users) and *pop_total* represents the total population of the electoral district. This transformation accounts for district size and reduces the heavy skewness of raw audience (see Figure A.3 in Appendix VIII.7.). I use the $\log(1 + x)$ form so that districts with zero recorded audience for a given outlet would not cause problem.

Because there are thousands of combinations for media outlets, time windows and user metrics, I used aggregation to simplify the model. I created a pro government media variable by averaging the log per capita values of specific sites. To maintain objectivity and avoid arbitrary judgments about political bias or influence, I relied on the (ATLO Team, 2022b) description of media ownership. This source identifies which outlets are financed by individuals or organizations with documented ties to the government. This method does not imply that these outlets are purely subjective or one sided. It's just an approximation that provides an objective data driven way to define the pro government media landscape.

I view these media data as recent reach instead of total exposure. This approach follows the literature on media effects, where recent viewership is a key driver of voter change (Gerber et al., 2011; Martin & Yurukoglu, 2017). Theoretically, the latest window (final 2 months before the election) is the most logical choice for capturing short term persuasion. For this reason I prioritize reach based metrics like *real_users* over intensity measures like page views or time per view. Furthermore, during the exploratory analysis, I examined the correlation between digital consumption and the government's vote share during the 2022 and 2024 elections. The *real_users* metric showed the strongest and most consistent correlation with the governmental margin (percentage point lead of the governmental block over the main opposition block), so it also serves as the most effective proxy for campaign momentum. Additionally, the pre-election consumption window closest to the vote showed no performance loss compared to earlier windows, confirming that these more recent audience signals are both reliable and representative of the final outcome (see Figure A.4 in Appendix VIII.8.).

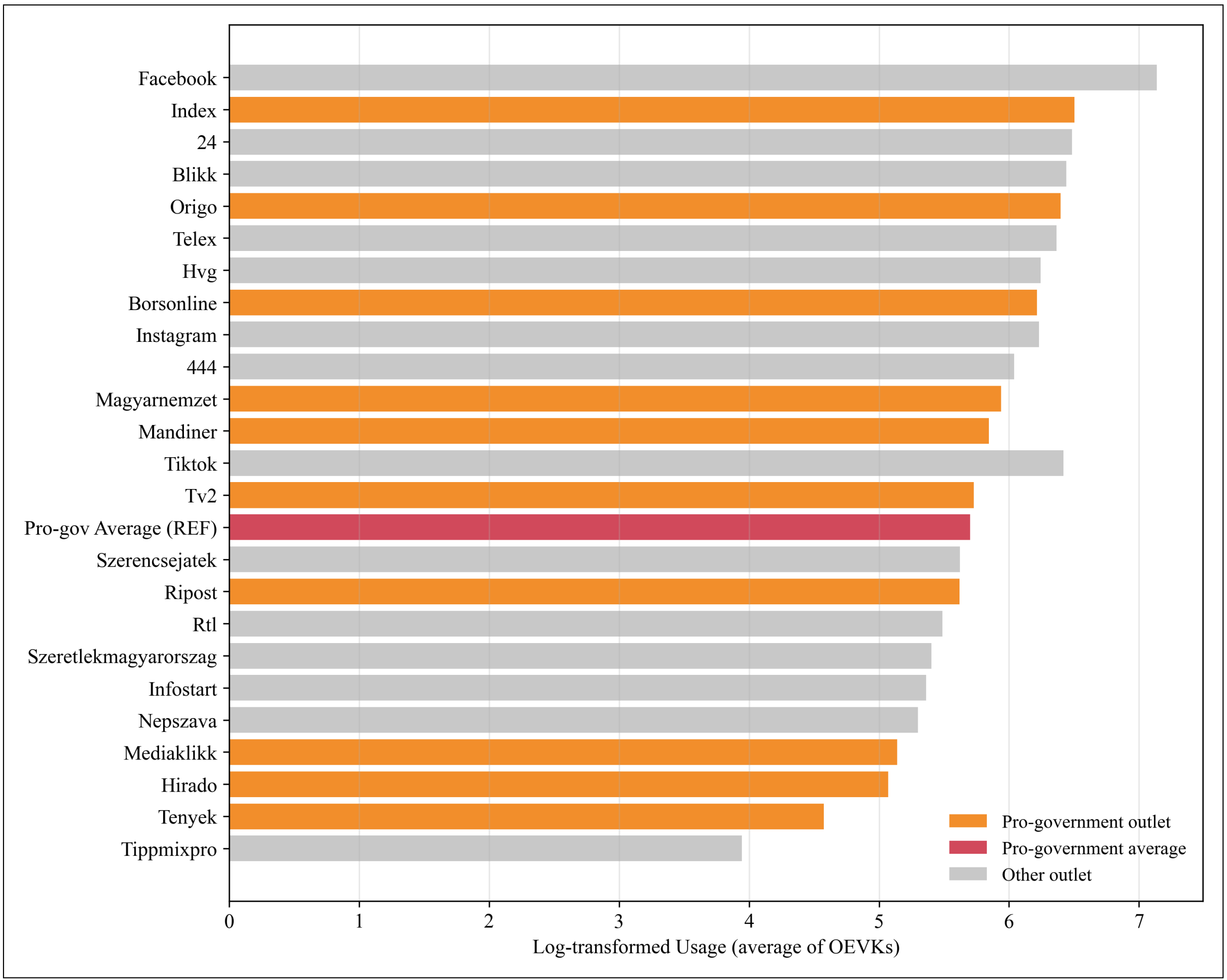

Figure 1: Media Outlet Usage in the Final Pre-election Window (2026)
Source: Compiled by the author based on Gemius Audience data. [5]

[5]Generative artificial intelligence was used for technical assistance in the preparation of this figure and its suggestions were reviewed and verified by the author.

The figure shows that media use in the final pre-election window is spread across a broad set of outlets, with most platforms exhibiting relatively similar levels of intensity. Pro-government outlets are well represented in the upper part of the ranking, which suggests that this media block has strong aggregate reach in the campaign period.[6] At the same time, the pro-government average is not an extreme outlier, indicating that its relevance comes from the combined presence of several outlets rather than from the dominance of a single channel.

IV. Methodology

This section describes the methodological framework used in the study. Instead of one single estimation I integrate several components: a national polling model, district-level fundamentals, local political leans and candidate-specific vote shares, all of which will be used to get the final seat simulation layer. This structure follows the basic logic of election forecasting where dynamic polling data is combined with long-term structural information and a rigorous treatment of uncertainty (Linzer, 2013; Stoetzer et al., 2019).

Because of Hungary's multi-party system, the vote shares have to be treated as compositional data, meaning, data describing the relative sizes of parts within a whole (Aitchison, 1986). In this setting, the shares must stay non-negative and they must sum to one. As Stoetzer et al. (2019) argues, addressing these constraints through log-ratio transformations is essential in multi-party forecasting, because it allows us to use the standard linear modeling techniques while ensuring that predictions are remaining within a mathematically valid range. For this reason, first I apply a very small minimum value $\varepsilon = 10^{-4}$ to the block shares where necessary, so that no individual share is exactly zero and the shares sum to one:

$$\tilde{p}_b = \frac{\max(p_b, \varepsilon)}{\sum_{k=1}^{B} \max(p_k, \varepsilon)}, \tag{2}$$

where p_b denotes the vote share of block b and p_k denotes the vote share of block k in the full set of blocks $k = 1, \ldots, B$.

After this step, the additive log-ratio (ALR) transformation is applied. This step converts the constrained vote shares into a set of unconstrained values, using the *other* block as the baseline:

$$z_b = \log\left(\frac{\tilde{p}_b}{\tilde{p}_{\text{other}}}\right), \qquad b \in \{\text{gov}, \text{opp}, \text{opp\_radical}\}. \tag{3}$$

[6] In this paragraph, I followed the stylistic and grammatical guidance provided by generative AI.

Later I will move back to normal vote share with the inverse transformation:

$$p_b = \frac{\exp(z_b)}{1 + \sum_{k \neq \text{other}} \exp(z_k)}, \qquad p_{\text{other}} = \frac{1}{1 + \sum_{k \neq \text{other}} \exp(z_k)}. \tag{4}$$

I have two linked targets. At national level, I will estimate the 2026 list vote shares by block and then turn them into seat distributions. At district level, I will estimate list shares, candidate shares, district winners and single-member district seats. This two-level structure is important because national vote shares alone do not determine district winners, while district structure alone does not tell us the current national campaign trend. The national and district parts therefore have to be estimated together and not as separate unrelated exercises (Erfort et al., 2026; Lauderdale et al., 2020). The model uses ALR space for estimation and simulation, but converts the results back to ordinary vote shares before calculating vote totals, district winners and parliamentary seats.

For model selection, parameter tuning and validation, I use a Leave-One-Election-Out (LOEO) approach. I train the model on past election data and test it against a single election that was kept out of the training set (Heidemanns et al., 2020; Stoetzer et al., 2019). This holdout election is excluded from every stage of the estimation and evaluation procedure, so all modeling choices are assessed on genuinely unseen electoral data. This ensures the model is calibrated using the actual historical differences between polling data and final election results.

IV.1. National Poll Model

IV.1.1. Simple Poll Benchmark

The first benchmark looks what the polls say on their own. For this, all polls from the final 60 days before election day are used to calculate a weighted average for each block based on sample sizes:

$$\hat{p}_b^{\text{simple}} = \frac{\sum_{i=1}^{n} w_i p_{ib}}{\sum_{i=1}^{n} w_i}, \qquad w_i = n_i, \tag{5}$$

where p_{ib} is the vote share of block b in poll i, $\hat{p}_b^{\text{simple}}$ is the weighted average and n_i is the sample size.

The standard deviation between polls and the "naive" standard error are also calculated:

$$SE_b^{\text{naive}} = \frac{s_b}{\sqrt{n}}. \tag{6}$$

Here, s_b represents the standard deviation across polls by block and n is the number of polls included for that bloc. This naive error shows how much the poll averages fluctuate, though it only captures sampling variation, so the noise from using a sample instead of the whole population (Shirani-Mehr et al., 2018).

In reality, poll error goes beyond simple sampling noise. Factors like pollster bias, turnout assumptions and partisan nonresponse can create larger "non-sampling" errors (Heidemanns et al., 2020; Shirani-Mehr et al., 2018). To account for this, the benchmark uncertainty is widened with a historical bias floor:

$$SE_b^{\text{final}} = \sqrt{\left(SE_b^{\text{naive}}\right)^2 + B_b^2}, \tag{7}$$

where B_b is a specific bias floor for each bloc. This floor represents the historical error found by subtracting poll averages from real results for each past election following the general idea of widening election day uncertainty with historical poll misses (Stoetzer et al., 2019). In my implementation, a leave-one-out rule is used, the bias floor for a certain election is the standard deviation of these misses from all other elections. This typical error acts as a minimum level of uncertainty, keeping the model realistic even when current polls look stable.

IV.1.2. Ridge Correction Benchmark

A second, more advanced benchmark uses a weighted Ridge regression model. This model learns how to correct current poll shares to better match final election-day results. Several controls are included to account for factors that influence accuracy: the number of days until the election, the share of undecided voters, the respondent base and categorical controls for pollster and interview mode.

Before estimation, numeric data is standardized and rare categories are grouped into an *Other* group. These controls help adjust for differences in when and how a poll was conducted. The model follows the standard Ridge formulation to stabilize the model when predictors are highly correlated (Hoerl & Kennard, 1970; Holland, 1973):

$$L(\beta_b) = \sum_{i=1}^{n} w_i \left(z_{e(i)b}^{\text{actual}} - x_i^\top \beta_b\right)^2 + \alpha \sum_{j=1}^{K} \beta_{bj}^2 \ , \tag{8}$$

note: all numerical variables were standardized

where $z_{e(i)b}^{\text{actual}}$ is the actual election day result in ALR space for block b in the election cycle of poll i and x_i is the vector of poll level information for poll i.

I chose the right penalty term (α) by running — the above mentioned LOEO — historical backtests instead of standard cross validation. This is because there are few independent datapoints, only four elections to learn from.

I also monitor the condition number of the model to ensure the system is robust. This number measures how much the forecast changes when the input data shifts slightly. A high condition number means the model is unstable and a small change in a poll could lead to a massive change in the forecast. Based on the leave one election out (LOEO) backtesting results, I will identify the optimal regularization parameter from historical cycles to use for the 2026 forecast. Then the penalty term is fixed ("hardcoded") for the final model. This constraint is necessary given the limited number of independent data points because it effectively reduces the model complexity to mitigate overfitting risks. I select a penalty parameter (α) that achieves a favorable balance between predictive error and condition number. This approach ensures the system remains robust and there is less change to overreact to stochastic noise in the polling data (Belsley & Klema, 1974).

IV.1.3. House Effects

Following the benchmarks, house effects are estimated, which is the consistent tendency of specific pollsters to over- or underestimate certain blocks (Heidemanns et al., 2020; Stoetzer et al., 2019). The error of each poll is modeled as:

$$e_{ib} = z_{ib}^{\text{poll}} - z_{e(i)b}^{\text{actual}} = x_i^{\top}\gamma_b + u_{c(i)b} + \varepsilon_{ib}, \qquad u_{c(i)b} \sim N(0, \sigma_{ub}^2). \tag{9}$$

Here z_{ib}^{poll} is the poll-based ALR value for block b in poll i and $z_{e(i)b}^{\text{actual}}$ is the actual election result in ALR space for block b in the election cycle to which poll i belongs. The vector x_i contains the same standardized predictors used in the Ridge model to account for polling conditions, except for the pollster. This is because it is modeled as the random effect $u_{c(i)b}$ which represents the unique bias of pollster c. Only this pollster-specific part is kept for the final correction. Time-based terms are not subtracted at this stage, as the later dynamic layer is designed to handle the actual movement of public opinion. The corrected poll share z_{ib}^{corr} is calculated by subtracting this random effect from the original poll result. These house effects are recalculated during each historical backtest using only data from other years. This ensures the model stays precise and does not see the result of the election it is trying to predict.

Following standard practice for multi-party models, this bias is assumed to follow a normal distribution. This allows for shrinkage which pulls the estimates of pollsters with few surveys toward the group average to prevent extreme corrections (Stoetzer et al., 2019).

IV.1.4. Random-Walk Poll Filter

Once corrected for house effects, individual polls are combined into a single daily average for each political block:

$$\bar{z}_{tb} = \frac{\sum_{i \in t} w_i z_{ib}^{\text{corr}}}{\sum_{i \in t} w_i}, \tag{10}$$

where z_{ib}^{corr} is the above mentioned house effect corrected share for poll i and w_i is its weight (sample size).

So the random walk filter is fitted to a daily series and not to separate poll rows. This prevents days with many published polls from affecting the filter multiple times just because more polls were released on that date (Linzer, 2013).

The random walk filter assumes that the true level of public opinion moves slightly from day to day, while polls provide noisy observations of this latent state (Linzer, 2013; Stoetzer et al., 2019):

$$m_{tb} = m_{t-1,b} + \eta_{tb}, \qquad \eta_{tb} \sim N(0, q_b), \tag{11}$$

$$\bar{z}_{tb} = m_{tb} + \nu_{tb}, \qquad \nu_{tb} \sim N(0, R_{tb}). \tag{12}$$

In this setup, m_{tb} is the latent support on day t. The term q_b is the evolution variance which shows how fast public opinion shifts. The term R_{tb} is the measurement variance and represents the noise in that day's polls. Both the daily changes in opinion and the poll errors are assumed to follow a normal distribution, because they represent random noise that is unpredictable and equally likely to be positive or negative (Linzer, 2013; Stoetzer et al., 2019).

The measurement variance R_{tb} is calculated for each day by summing the variance between different polls with a penalty for small sample sizes. The penalty is based on the ratio of the median daily total of respondents to the current daily total. This ensures that days with inconsistent results or few respondents have a higher level of uncertainty.

Using the notation above, this can be written as:

$$R_{tb} = s_{tb}^2 + c \cdot \frac{\tilde{N}}{N_t}, \tag{13}$$

where s_{tb}^2 denotes the within day poll variance for block b, N_t is the total respondent base on day t, $\tilde{N}$ is the median daily respondent base in the estimation window and c is a small fixed penalty constant. It is fixed in advance as a conservative scaling factor so that days with fewer respondents are treated as less precise without allowing the penalty term to dominate the within day poll variance.

The evolution variance q_b is derived from the daily volatility of the poll series. Actual shifts in public opinion are separated from random sampling noise by subtracting the estimated measurement error from the total variation, which follows the logic of partitioning polling error into process and measurement components (Linzer, 2013; Stoetzer et al., 2019).

On each day, the estimate from the previous day is first carried forward and its uncertainty is increased by the evolution variance q_b.[7] If a poll observation is available on that day, the model updates this prediction toward the observed daily mean. The size of this update depends on the relative size of the prediction uncertainty and the measurement variance R_{tb}, so precise poll signals move the latent path more strongly while noisy poll signals move it only slightly. If no poll is available on day t, no update is made and the model simply carries the previous prediction forward with higher uncertainty. This is important because support may still evolve between poll releases.

This house effects random walk poll model is essentially a Kalman-style updating procedure (Linzer, 2013; Stoetzer et al., 2019) that formalizes the balance between trend persistence and new polling evidence. By tracking the latent support path m_{tb} and its uncertainty, the filter smooths the daily polling series through the weighted adjustments described above.[7] This ensures the forecast remains robust to short-term noise while detecting genuine shifts in voter sentiment as the campaign progresses.

At this stage each block is still estimated independently. Relationships between different blocks are added later when the polls are combined with the fundamentals layer.

IV.2. District Fundamentals and Local Lean

IV.2.1. Feature Construction

Complementing the national poll model, the next step is the district-level fundamentals. These fundamentals represent a slow-moving baseline that is especially influential early in the campaign, before polling signals begin to dominate the forecast (Heidemanns et al., 2020; Stoetzer et al., 2019). The feature set are the variables discussed in the data section (see Section III.) like demography, labor market, fiscal health and media reach (Hanretty et al., 2018).

The estimation follows a "staircase" logic to capture information moving at different speeds:

1. *Socio-economic Baseline:* Long-term structural data (education, employment etc...) that defines the character of a district (Hanretty et al., 2018; Stoetzer et al., 2019).

2. *Historical Anchor (Lags):* Results from previous elections. These lags capture the latent partisanship and historical persistence of voting patterns that demographics alone might miss (Lauderdale et al., 2020; Linzer, 2013).

3. *Dynamic Signal:* The previously discussed pro-government digital media reach, which is intended to capture short term campaign dynamics (ATLO Team, 2022b; Gerber et al., 2011).

[7] In this paragraph, I followed the stylistic and grammatical guidance provided by generative AI.

But first, I perform structural feature selection for the socioeconomic indicators. I define four nested candidate sets to identify the most robust baseline configuration:

1. *Demography*: Population logs and age-group shares (young/old) to capture the basic electoral landscape.

2. *Core economy*: Labor market indicators (total and university-educated unemployment) and local fiscal health (budget balances).

3. *Modernization*: Wealth and infrastructure proxies, specifically car ownership and recent vehicle registration rates.

4. *Public services and housing*: Healthcare capacity (doctors per capita) and the intensity of private residential construction.

The optimal structural layer is selected via LOEO loop based on national Mean Absolute Error (aggregating all district-level predictions into a single national projection). Once this foundation is chosen, it is combined with the historical anchors and dynamic signals to form the final predictor vector.

IV.2.2. National Fundamentals Prior From District Models

The national fundamentals prior is constructed using a bottom-up approach. Rather than estimating a national projection directly, vote shares are first predicted for each individual district and subsequently aggregated (Erfort et al., 2026; Lauderdale et al., 2020). This ensures that the national starting point is grounded in the specific demographic and economic profile of each local area.

The first considered model is a random intercept model, where it predicts results at the district level, although it incorporates a county-level random intercept ($u_{\text{county}(d),b}$). This is because districts within the same county often share unobserved regional characteristics, local political traditions or economic trends that are not fully captured by district-level variables alone (Hanretty et al., 2018; Lauderdale et al., 2020). For each district d, election e and block b, the following specification is estimated:

$$z_{deb} = x_{de}^{\top}\beta_b + u_{\text{county}(d),b} + \varepsilon_{deb}, \tag{14}$$

where z_{deb} is the observed vote share in district d during election e (in ALR space), x_{de} contains the chosen district-level (standardized) structural variables, $u_{\text{county}(d),b}$ is the county-level random intercept representing regional variation and ε_{deb} is the idiosyncratic error term representing district-specific noise (Hanretty et al., 2018; Lauderdale et al., 2020).
However, similar to the poll correction methodology described earlier, Ridge regression can also be an effective tool for handling high correlation between demographic predictors (Hoerl

& Kennard, 1970; Holland, 1973). The Ridge coefficients are estimated by minimizing the following penalized loss function:

$$L(\beta_b) = \sum_{d,e} (z_{deb} - x_{de}^{\top}\beta_b)^2 + \alpha \sum_{j=1}^{K} \beta_{bj}^2 \tag{15}$$

where z_{deb} and x_{de} are the same as above, β_b represents the block-specific coefficients and α is the penalty parameter that shrinks the coefficients toward zero. This method is particularly useful when predictors such as population, unemployment and doctors are highly correlated.

To choose the functional form for the national fundamentals prior, I use Leave One Election Out backtesting over the historical elections. I compare the Ridge and county level random intercept specifications on that same selected predictor set. In this prior layer, the main selection criterion is national Mean Absolute Error (MAE) after the district predictions are aggregated to national level, while district level MAE is kept as a secondary diagnostic. The Ridge penalty α is evaluated under the same historical LOEO logic and then fixed for the final forecast. If one specification shows a clear holdout advantage, that specification will be used. If the two forms perform very similarly, Ridge will be preferred on parsimony grounds, because it gives a simpler and more transparent baseline and does not require the estimation of an additional county variance component from a very small election history.

Once district-level shares are predicted, they are combined into a single national prior:

$$\hat{p}_{eb}^{\text{prior,nat}} = \frac{\sum_{d=1}^{D} w_{de}\hat{p}_{deb}^{\text{prior}}}{\sum_{d=1}^{D} w_{de}}. \tag{16}$$

Here, $\hat{p}_{deb}^{\text{prior}}$ denotes the predicted prior vote share of block b in district d for election e, while $\hat{p}_{eb}^{\text{prior,nat}}$ denotes the corresponding national prior vote share of block b in election e. The weight w_{de} is the number of valid list votes in district d for election e. This approach ensures the national starting point is tied to the real district structure of the country, reflecting the idea that fundamentals provide a baseline before polls take over (Heidemanns et al., 2020; Stoetzer et al., 2019).

For the 2026 forecast, since current district valid list votes are not yet observed, I proxy the aggregation weights with the last observed district level valid list votes from the previous parliamentary election.

IV.2.3. District Lean Model

The next step involves estimating the district lean, which is the natural difference between a specific district and the national average (Kiewiet de Jonge et al., 2018; Lauderdale et al., 2020).

This lean is defined in ALR space as:

$$\Delta_{deb} = z_{deb}^{\text{list}} - z_{eb}^{\text{nat}}, \tag{17}$$

where z_{deb}^{list} represents the log-ratio (ALR) of the list vote share for block b in district d during election e and z_{eb}^{nat} denotes the national average log-ratio for the same block and election.

Similar to the national fundamentals prior, I evaluate two base specifications and take forward the one that performs better in Leave One Election Out backtesting:

- *Random-Intercept Model:* A multilevel specification that accounts for regional variation through a county-level random effect ($v_{\text{county}(d),b}$):

$$\Delta_{deb} = x_{de}^{\top}\beta_b + v_{\text{county}(d),b} + \xi_{deb}. \tag{18}$$

- *Ridge Regression:* A penalized linear model that handles collinearity between predictors by shrinking coefficients:

$$L(\beta_b) = \sum_{d,e}(\Delta_{deb} - x_{de}^{\top}\beta_b)^2 + \alpha \sum_{j=1}^{K} \beta_{bj}^2, \tag{19}$$

  where x_{de} is the vector of standardized structural features, β_b represents the block-specific coefficients and α is the penalty parameter which will be selected through the same procedure described earlier.

This design follows the "partisan lean" logic used in international forecasting, where local units are modeled relative to the national race rather than in isolation (Kiewiet de Jonge et al., 2018; Lauderdale et al., 2020). This approach has also been applied in practice, most notably in the YouGov constituency model (YouGov, 2017).

After the optimal β_b coefficients are found and applied to the 2026 data, the model first produces a base district lean forecast, denoted by $\hat{\Delta}_{db}^{\text{base}}$. This is the slow structural prediction before any recent correction is added.

The base lean model starts from the previously selected socio economic baseline features. In the LOEO implementation, this set is checked again within each training fold so only those predictors are kept that have sufficient historical coverage.

The base lean model does not include lagged lean values or media features. This is because these variables have more limited historical coverage than the core structural predictors. After the better base specification is chosen, the recent correction layer then can add recent lagged lean values, recent media features or both, depending on the correction specification and the available data in the relevant holdout fold.

IV.2.4. Recent Correction Layer

Last but not least, an additional recent correction layer is added to the base district lean model. The goal is to capture district level shifts that the slower structural model may miss, such as recent political momentum or newly relevant media patterns. This follows the general idea of time varying adjustment in Linzer (2013) and Heidemanns et al. (2020), but here it is implemented in a simpler way as a separate residual correction on top of the base lean model (Heidemanns et al., 2020).

For this only the recent elections are used, which is 2022 and 2024. I define a correction target in ALR space as:

$$r_{deb} = \Delta_{deb} - \hat{\Delta}^{\text{base}}_{deb}, \tag{20}$$

where r_{deb} is the recent correction target for district d, election e and block b, Δ_{deb} is the observed district lean and $\hat{\Delta}^{\text{base}}_{deb}$ is the lean predicted by the base model.

Here only ridge model is used because the training sample is very limited. The random intercept model would require a much larger training sample to reliably estimate the county-level effects, while the Ridge regression can still provide a useful correction by shrinking coefficients toward zero when the data is limited. In other words, while a random intercept model might be more appropriate for the broader district lean estimation where more historical data is available. In that setting, Ridge regression is the safer choice, because it can shrink a possibly unstable set of lag and media coefficients toward zero, while a county level random intercept would add another layer of complexity on a much smaller training window.

For each block b, the correction model is estimated as:

$$L(\gamma_b) = \sum_{(d,e)\in\mathcal{R}} w_{de} \left(r_{deb} - x^{\text{corr}\top}_{de}\gamma_b\right)^2 + \alpha \sum_{j=1}^{K} \gamma^2_{bj}, \tag{21}$$

note: all numerical variables were standardized

where $\mathcal{R}$ contains the recent elections of 2022 and 2024, w_{de} is the district weight based on valid list votes, γ_b is the vector of correction coefficients for block b, α is the penalty parameter and x^{corr}_{de} denote the vector of correction features, which may contain lagged lean variables, screened recent media variables or both.

The predicted recent correction is then

$$\hat{r}_{deb} = x^{\text{corr}\top}_{de}\hat{\gamma}_b. \tag{22}$$

Because this step is estimated only on the recent elections of 2022 and 2024, it should be read as a cautious adjustment to the base model rather than as a fully independent forecasting test.

The final district lean forecast is then updated as

$$\hat{\Delta}_{db}^{\text{final}} = \hat{\Delta}_{db}^{\text{base}} + \lambda \hat{r}_{db}, \tag{23}$$

where $\hat{r}_{db}$ is the predicted recent correction term and $\lambda \in [0, 1]$ is a weight, which controls how strongly the recent correction is applied. If the selected weight is 0, there is no recent correction. This allows the model to react to recent district specific movement while keeping the long run structural prediction as the main anchor.

After this correction, the district level support for the 2026 election is reconstructed as

$$\hat{z}_{db}^{\text{list}} = \hat{z}_{b}^{\text{nat}} + \hat{\Delta}_{db}^{\text{final}} + \kappa_b. \tag{24}$$

Here $\hat{z}_{b}^{\text{nat}}$ denotes the national-level ALR forecast for block b, while κ_b is an adjustment term that ensures the district forecasts aggregate exactly to that national forecast.

This alignment step is crucial for maintaining consistency between the national and local levels (Erfort et al., 2026; Lauderdale et al., 2020).

The initial plan was to consider three potential specifications for the district level correction layer. These include one using lagged lean variables, one using screened recent media variables and one combining both. Each specification is evaluated using grouped county holdouts. In this design, whole counties are held out from the estimation to ensure validation is performed on entirely unseen geographic areas. This approach reduces geographic leakage, because districts within the same county often share similar political and media patterns. It provides a stricter test than random district level holdouts.

However, with only two recent elections available, which are 2022 and 2024, the initial plan would train on a single election and validate on another. This is not a statistically reliable basis for model selection. Optimizing a free parameter on such a limited sample carries a high risk of overfitting to idiosyncratic noise rather than capturing a generalizable relationship. I instead fix the correction feature set to include both lagged values, media reach features and the correction weight $\lambda = 1.0$ ex ante.

The decision to retain both lagged lean values and media reach features, despite their performance in the holdout tests, is rooted in theoretical robustness. Lagged results provide the necessary "Historical Anchor," which is a standard requirement for capturing stable local partisanship that demographic variables alone often understate (Linzer, 2013). The inclusion of digital media reach as a "Dynamic Signal" is specifically justified by the documented pro-government concentration of the Hungarian media landscape, which acts as a proxy for localized campaign intensity (ATLO Team, 2022b).

Applying the full correction weight ($\lambda = 1.0$) is the most transparent and conservative approach under data constraints. Any attempt to tune λ to a value between 0 and 1 would essentially be an exercise in "fitting the noise" of the 2022 and 2024 cycles. By fixing λ at unity, I ensure that the model's predicted recent shifts are fully incorporated into the forecast without introducing a manually adjusted "dampening" factor that cannot be empirically validated.

IV.3. Forecast Integration, Candidate Effects and Seat Simulation

IV.3.1. Combining Polls And Fundamentals

After estimating the national poll forecast and the national fundamentals prior, they are combined into a single national forecast. This follows standard practice where fundamentals provide a baseline and polls update that baseline as election day approaches (Linzer, 2013; Stoetzer et al., 2019). There are three different common rules that can be used for mixing these sources:

- *Fixed weight combination:* A common heuristic where current polling is given a $x\%$ weight and fundamentals a $y\%$ weight. This provides a transparent, simple baseline that prioritizes recent sentiment while maintaining a significant "anchor" in structural reality (Heidemanns et al., 2020; Kayser et al., 2022).

- *Inverse-variance approach:* This rule assigns weights based on the historical reliability of each source. In additive log ratio (ALR) space, the fundamentals prior and the poll forecast receives a weight based on the inverse of its historical error variance. So the model gives more weight to the source that had smaller forecast errors in past elections (Lock & Gelman, 2010; Stoetzer et al., 2019).

- *Empirical convex combination:* This rule identifies the optimal balance by testing a range of weights against historical data (Linzer, 2013; Lock & Gelman, 2010).

The final national forecast is produced by combining the poll based forecast and the fundamentals prior using a fixed 70 to 30 weighted average. This simple and transparent rule ensures that the final result is strictly bounded between the two estimates.

$$\hat{p}_{eb}^{\text{comb}} = \omega \hat{p}_{eb}^{\text{poll}} + (1 - \omega) \hat{p}_{eb}^{\text{prior,nat}}. \tag{25}$$

Here $\hat{p}_{eb}^{\text{poll}}$ denotes the national poll forecast for block b in election e, $\hat{p}_{eb}^{\text{prior,nat}}$ denotes the aggregated national fundamentals prior, $\hat{p}_{eb}^{\text{comb}}$ is the resulting combined forecast and ω is the fixed weight of 70%.

While an empirical grid search is a theoretical alternative, I adopt this fixed weight for the final campaign period, because with only four elections available for Leave One Election Out

(LOEO) validation, optimizing ω would be statistically unreliable. Relying on an optimized weight from such a small dataset carries a high risk of overfitting to specific past anomalies rather than identifying a robust and generalizable signal.

This split is a transparent practical benchmark for the final campaign period. It gives primary weight to polls while retaining a structural anchor from the fundamentals prior (Brown & Chappell, 1999; The Economist, 2020). Using a high weight for polls ensures that recent voter sentiment is the primary driver of the forecast while the fundamentals prior remains as a necessary structural anchor. This balance acknowledges the high informational value of polls during the final campaign period and the fundamentals provide a baseline.

IV.3.2. National Uncertainty

While the point forecast provides a central estimate, the model must also account for national-level uncertainty. In a multi-party election, uncertainty cannot be modeled for each block in isolation. Vote share errors across all blocks must sum to zero, so an overestimate on one block is distributed across the remaining blocks (Aitchison, 1986). However, this does not force every pairwise correlation to be negative. In ALR space, where each block is expressed relative to the same baseline, two blocks can have positively correlated errors if both move jointly relative to that baseline. Therefore, simulating realistic election scenarios requires estimating the historical correlation structure of prediction errors between different political blocks in ALR space (Stoetzer et al., 2019).

However, measuring these relationships accurately is difficult because data is only available from few past elections. If the model relies entirely on this small sample, it might mistake random coincidences for strong, permanent patterns. To prevent this, a statistical shrinkage technique is applied (inspired by the Bayesian regularization method, (Ledoit & Wolf, 2004; Schäfer & Strimmer, 2005; Stoetzer et al., 2019)).

This technique calculates a weighted average between the raw empirical correlation matrix and an identity structure (representing a baseline of zero correlation).[8] This pulls the estimated correlations away from potentially spurious extremes toward a more conservative independent state. This approach preserves the directional dependencies observed between political blocks while preventing the simulation from becoming overly sensitive to unstable empirical estimates. This adjusted covariance matrix is then used in the election simulations.

So the model starts from the historical covariance matrix, but it does not treat that matrix as fully trustworthy when it was estimated from only a handful of elections. Instead, it partially pulls the estimate toward a zero correlation benchmark. This means that clear and repeated historical co-movement can still remain in the matrix, but weak patterns are treated more cautiously. In

[8] In this paragraph, I followed the stylistic and grammatical guidance provided by generative AI.

this way, shrinkage reduces the chance that the simulation learns an exaggerated dependence structure from a very small sample.

The weight is based on how many past elections are available for estimating this national covariance matrix, relative to a fixed "prior strength" parameter that acts as a skepticism factor. If only a few past elections are available (for backtest), the model shrinks the estimated correlations more strongly toward zero. This shrinkage also becomes stronger when the prior strength value is higher, which makes the model more skeptical and improves numerical stability. A lower prior strength value lets the observed historical correlation patterns play a larger role (Gelman, 2006; Stoetzer et al., 2019).

In the implementation, this parameter is fixed rather than being tuned separately, since the historical sample is too small for reliable calibration. This follows the basic logic of shrinkage, where with only a few past elections, the estimated correlations are more likely to reflect random noise than stable historical patterns (Ledoit & Wolf, 2004; Schäfer & Strimmer, 2005). I therefore set the prior strength conservatively, so the covariance step is less affected by unstable correlation estimates and the simulation does not produce overly narrow uncertainty bands.

Concretely, I set the prior strength to 6.0. Since there are 4 available elections, this gives the data a weight of $n/(n+k) = 4/(4+6) = 0.4$ and the identity prior a weight of 0.6. In other words, the prior is treated as equivalent to 1.5 times the observed sample, so the data receives less than half the total weight.

At this sample size, clear and repeated patterns in the error correlations should survive the regularisation, while weaker or one-off correlations are absorbed into the identity. As a second protection, any pairwise correlation that remains above 0.45[9] after shrinkage is manually capped at that value. This prevents two blocks from appearing so tightly linked that an error in one nearly forces a matching error in the other, which would make the simulation outcomes unrealistically concentrated. Together, these two choices ensure the simulation remains tractable and does not overfit to the specific correlation patterns of a small number of past elections (Ledoit & Wolf, 2004; Schäfer & Strimmer, 2005).

IV.3.3. Candidate-Gap Model For Single-Member Districts

In the Hungarian system, a block's support in a specific district is not always the same as its national list support (Erfort et al., 2026; Kovacs & Vida, 2015). To account for this, I define a candidate-gap layer in additive log-ratio (ALR) space as the difference between the district candidate result and the district list result for block b in district d and election e:

$$g_{deb} = z_{deb}^{\text{cand}} - z_{deb}^{\text{list}}. \tag{26}$$

[9] I chose this threshold as a cautious middle ground, permissive enough to leave moderate correlations intact, but strict enough to rule out implausibly strong cross block dependence in such a small sample.

The model predicts this gap directly rather than predicting list and candidate shares as two independent targets. This prevents errors from two separate models from compounding and ensures that the candidate prediction remains logically anchored to the party's general popularity in that district.

Consistent with the national prior and district lean models described previously, I use a *hierarchical mixed-effects regression* with county-level random intercepts to predict the 2026 gap using the following standardized signals (Hanretty et al., 2018; Lauderdale et al., 2020):

1. *Historical Gaps:* How much a party's candidate outperformed or underperformed their list in 2018 and 2022.

2. *Structural Variables:* The same set of socio-economic indicators used in the fundamentals and lean models, which here help identify local conditions.

3. *Current List Support:* The predicted 2026 list share, which acts as the anchor for the candidate's performance.

The model is compared against a simpler benchmark, where the candidate share is assumed to equal the list share (a zero gap). The version that minimizes the prediction error in a grouped county holdout — an out-of-sample test where the model must predict results for districts in counties it did not use during training — is then selected for the final forecast.

Because only one full transition cycle (2018 to 2022) is available for training, the predicted gap is regularized using the same statistical shrinkage logic described earlier. This ensures the model defaults to the party list unless there is strong empirical evidence for a candidate-specific effect.

Specifically I set the candidate-gap covariance shrinkage prior strength to 25.0 and the post-shrinkage correlation cap to 0.35, chosen by the same $n/(n+k)$ logic as the national layer. However, the apparent sample contains 106 districts, yet all of them come from a single parliamentary transition from 2018 to 2022. The effective number of independent cycles is therefore 1, not 106, because districts within the same election share the same election-level shocks and are not independent draws for covariance estimation (Ledoit & Wolf, 2004). Setting the prior strength to 25.0 gives the data a weight of $n/(n+k) = 1/(1+25) \approx 0.04$ and the identity prior a weight of 0.96, which means candidate patterns from one cycle receive very little influence over the estimated covariance structure. This is needed, since learning from only one transition is difficult (Ledoit & Wolf, 2004; Schäfer & Strimmer, 2005).

The 0.35 correlation cap is set lower than the 0.45 used at national level because candidate-specific effects are a secondary adjustment around the list forecast. Allowing higher cross-block correlations in this layer would generate unrealistically synchronized candidate shocks across districts (Stoetzer et al., 2019). A cap of 0.35 still preserves moderate co-movement when the data supports it.

Regarding the choice of architecture, while Ridge regression is preferred for the broader structural layers, the candidate gap model represents an exception where hierarchical mixed effects provide superior stability (Hanretty et al., 2018; Selb & Munzert, 2011).[10] This choice is driven by the territorial nature of local campaigning in Hungary, where candidate specific effects often cluster within counties due to regional party organizational strength. This approach explicitly partitions the variance between regional trends and district specific noise, allowing for a more nuanced form of regularization. By pulling noisy district estimates toward a stable county level mean, the architecture ensures that candidate deviations are only preserved when they represent consistent regional patterns rather than stochastic outliers. Consequently, this estimator is chosen to handle the spatial dependencies inherent in the SMD layer.

IV.3.4. Seat Allocation

District-level list and candidate shares are translated into parliamentary seats using a block-level approximation of the current Hungarian rules. This is a mixed system with 106 single-member districts and 93 national-list seats (National Election Office of Hungary, 2026b). I applied a simplification here by not accounting for potential national minority mandates.

A key feature of this system is compensation votes. The model follows these rules:

- If a block *loses* a district, all of its candidate votes are added to its national list as compensation.
- If a block *wins* a district, only the surplus votes — those beyond what was needed to beat the second-place candidate — are added:

$$C_{db} = \begin{cases} v_{db}^{\text{cand}}, & \text{if block } b \text{ does not win district } d, \\ \max\left(v_{db}^{\text{cand}} - v_{d(2)}^{\text{cand}} - 1, 0\right), & \text{if block } b \text{ wins district } d. \end{cases} \tag{27}$$

where C_{db} denotes the compensation votes for block b in district d, v_{db}^{cand} represents the total votes received by the block's candidate and $v_{d(2)}^{\text{cand}}$ refers to the vote count of the second-place candidate in that same district. The term -1 means that the winner needed one vote more than the second place candidate, so only votes above that minimum winning margin count as surplus compensation. The $\max(\cdot, 0)$ term ensures that compensation cannot become negative if the simulated or rounded vote totals make this surplus expression slightly smaller than zero. These compensation votes are added to the national list votes and the D'Hondt rule is used to allocate the 93 list seats while respecting the 5% legal threshold. This is a proportional divisor method in which blocks with more votes receive more list seats, with seats allocated step by step based

[10] In this paragraph, I followed the stylistic and grammatical guidance provided by generative AI.

on the vote totals (National Election Office of Hungary, 2026b). The final seat count for each block is the sum of its district winners and its list seats.

However, the *Other* block cannot be handled correctly in this setup, because it is only a residual vote category that combines several small parties and independents, while the D'Hondt calculation treats it as if it were one unified party. This makes it much easier for the block to pass the threshold than in reality. For this reason, I exclude the *Other* block from the final list seat allocation and calculate seats only for the government, opposition and radical opposition blocks. The *Other* share is still kept in the vote forecast so that the full vote composition remains complete.

IV.3.5. Monte Carlo Simulation

The final output is not a single point estimate, but a distribution of possible outcomes (Linzer, 2013; Stoetzer et al., 2019). To generate this distribution, the model runs 1,500 Monte Carlo iterations, following the simulation based approach used by Linzer (2013) and The Economist (2020). This number of iterations is sufficient enough, since increasing the number of draws would not produce any meaningful gain in precision.

While the point forecast uses the combined national forecast, the predicted district lean and the predicted candidate gap, the simulation model adds uncertainty around these central predictions. At national level, this uncertainty is based on the adjusted covariance matrix of past national forecast errors in ALR space described above. At district level, it is based on past residuals from the district lean model, which is the difference between the observed district lean and the fitted district lean in historical data. On the single member district side, it is based on past candidate gap residuals, that is, the difference between the observed candidate gap and the candidate gap implied by the selected central single member district specification in historical data, either the separate candidate gap model or the zero gap benchmark.

For each draw, the model identifies district winners, calculates list seats and determines total seat counts. By aggregating these $1,500$ results, the model generates a comprehensive distribution of possible outcomes, allowing for the estimation of event probabilities (such as a block winning a majority). These are interpreted not as exact real-world probabilities, but as "scenario frequencies" that clarify the range of outcomes supported by the model's assumptions (Linzer, 2013; Stoetzer et al., 2019).

IV.4. Validation Strategy

IV.4.1. Validation Logic

I use different validation rules because the national and district layers face different risks. At national level the main question is temporal. A national poll model or national prior is only useful if it can learn from past elections to predict a new one and temporal holdout matches that problem directly. However, at district level there is an extra spatial risk. Districts in the same county often share the same political, social and media patterns. If very similar districts appear in both the training and the test set, then the reported accuracy can become too optimistic because there are geographic information leaks across the split (Hanretty et al., 2018; Lauderdale et al., 2020).

For this reason, I use a layered validation design. Temporal holdout is the main rule when a layer has to learn from earlier elections and predict a later one, while grouped county holdout is used when the main danger is geographic leakage between similar districts.

IV.4.2. Application To The Layers

For the national poll model, the aggregated fundamentals prior and the poll and prior combination step, I use temporal backtesting over the four historical elections in the sample. This is more appropriate than a random split because the real task is to predict a future election and not an another observation from the same campaign period (Linzer, 2013; Stoetzer et al., 2019). National forecasts are judged mainly by Mean Absolute Error (MAE). For the national poll model, Root Mean Squared Error ($RMSE$) is also reported as a supplementary descriptive measure, but model choice is based on MAE.

At district level, the broader structural layers — the base district fundamentals prior and the base district lean model — also use election holdouts because they must generalize across election cycles. However, temporal holdout alone is not enough for the narrower local correction layers. Since the recent correction step and the candidate gap model are estimated on much shorter training window (because the data is more limited), the main risk there is geographic leakage rather than temporal overfitting. For these layers I therefore use grouped county holdout, where whole counties are excluded from estimation and used only for validation.

District forecasts are judged mainly by district level MAE. For the district lean model I also report winner hit rate as a secondary measure because parliamentary seat conversion depends strongly on which block finishes first in each district (National Election Office of Hungary, 2026b). In the Hungarian system, district wins convert directly into parliamentary seats and the margin of victory only affects the size of the compensation pool, making winner accuracy a stronger indicator of forecast quality for seat allocation than the raw error average. For the candidate gap layer, the grouped county holdout comparison is made against a simpler list-proxy benchmark where candidate share is assumed to equal list share.

IV.4.3. Scope And Limits

The validation design is layered rather than fully uniform. The national layers and the broader district structure layers are judged mainly by temporal election holdouts, while the narrower local correction layers are judged with grouped county holdout to reduce geographic leakage. The seat allocation block is not treated as a separate forecasting model. Instead, it is checked by reconstructing historical parliamentary seat totals under the electoral rules. Because only a few elections are available, this validation can reject clearly weak specifications but cannot justify very fine distinctions between similar models. Final simulated probabilities are therefore interpreted as scenario frequencies conditional on the model assumptions rather than as exact real world probabilities (Linzer, 2013; Stoetzer et al., 2019).

The main practical limitation is that some later layers rest on a much narrower evidence base than the core national and district structure models. Comparable Gemius media data is available only for the 2022 parliamentary election and the 2024 European Parliament election, so the media based correction step has only 50% backtest coverage across the four holdout elections and is treated as a recent additional signal rather than as a core predictor that is already validated across many cycles. The candidate gap layer is even narrower, because it is trained on only one full parliamentary transition, from 2018 to 2022. For this reason the candidate effect is regularized strongly and pulled toward the simpler list share proxy when the signal is weak. The final seat check is also limited because it validates a block level approximation of the Hungarian electoral rules, not a full party alliance legal simulation.

A complete summary of all model and parameter selection decisions is provided in Appendix VIII.12..

V. Results

V.1. National Poll Model Results

V.1.1. Simple Poll Benchmark

The simple weighted average, computed over polls within the final 60 days before each election, produced a mean absolute error of 1.27 percentage points across the four historical holdout elections. From Table 4 it can be seen, that the lowest single-election error was 0.91 percentage points for 2018 and the highest was 1.66 points for 2022.

Election	MAE (pp)	Between-poll SD (pp)	Naive SE (pp)	SE with bias floor (pp)
2018 parliamentary	0.91	3.66	0.50	1.67
2019 EP	0.97	2.89	0.63	1.53
2022 parliamentary	1.66	4.20	1.40	1.85
2024 EP	1.53	3.55	0.59	1.13
Mean	1.27	3.58	0.78	1.55

Table 4: Simple poll benchmark results by historical election

Source: Compiled by the author.

note: pp means percentage points

The spread between individual polls in every election was considerably larger than what sample size alone would predict. This means that systematic bias, not just sampling noise, drives the total uncertainty, which is why the historical bias floor is included in the benchmark interval. Adding the bias floor widened the adjusted standard errors substantially in all elections.

Using the same four election backtest, the interval with the bias floor contained the true result in 62.5% of block election combinations, compared to 50% for the version without the bias floor. Both the point estimates are identical across these two variants, because the bias floor only widens the interval and does not shift the central forecast. Both coverage figures are far from full calibration, but the bias floor is clearly the safer choice for reported uncertainty.

V.1.2. Ridge Correction Benchmark

The Ridge correction benchmark produced a mean absolute error of 5.95 percentage points, which is more than four times worse than the simple average. The penalty parameter α was set to 20 based on the condition number criterion. This value kept the system well-conditioned while the MAE sensitivity to alpha was negligible across the entire tested range. Since this model performs clearly worse, I will explain the alpha selection later in section where regularisation plays a more meaningful role.

V.1.3. House Effects

Table 5 shows the estimated house effects for the pollsters with sufficient observations. The estimates are reported in additive log-ratio (ALR) space (because this is where the whole model operates). Here a positive value means the pollster tends to report higher support for that block than the estimated latent state and a negative value means they tend to report lower support. Because ALR expresses each party block as a log-ratio against the *Other* category, the same numerical bias will look bigger or smaller in percentage-point terms depending on where all the other blocks sit at that moment (a shift of one ALR units means something different when a party polls at 20% than when it polls at 40%). Expressing the bias in log-ratio units avoids this ambiguity and states exactly what the model estimated and corrected for. The model already controls for the *Összes* vs *Biztos* methodological difference, so the displayed values represent the pollster-specific residual after this covariate has been accounted for.

Pollster	Government	Opposition	Radical opposition	Avg. abs. effect
Tárki	+0.79	+0.43	+0.64	0.62
Századvég	−0.49	−0.53	−0.43	0.48
Publicus	+0.25	+0.44	−0.57	0.42
Psyma	+0.39	+0.40	+0.28	0.36
Nézőpont	−0.38	−0.38	−0.16	0.31
ZRI	−0.48	−0.14	−0.30	0.31
e-benchmark	+0.08	−0.14	−0.24	0.15
Iránytű	+0.02	+0.01	+0.40	0.14
Civitas	+0.11	+0.06	+0.27	0.14
Medián	−0.20	−0.18	+0.05	0.14
Republikon	−0.04	+0.10	−0.05	0.06
Alapjogokért Központ	−0.04	−0.03	+0.06	0.04

Table 5: Estimated house effects by pollster (ALR space)
Source: Compiled by the author.

The spread in estimated house effects is wide enough to meaningfully distort the national forecast if left uncorrected. Pollsters broadly split into two groups, some consistently shift party shares upward relative to the baseline, while others deflate them. The same raw deviation carries a different weight depending on the overall composition of support at that moment. What matters practically is that the divergence is large enough and the pollsters directionally opposed enough, that the unweighted average of published polls is sensitive to which firms happened to be active in any given window. Correcting for house effects before aggregation removes this dependency.

V.1.4. Random-Walk Poll Filter

The random-walk filter is applied on top of the house effect correction. In the outer[11] LOEO holdout comparison, the house effects plus random-walk path performed considerably worse than the simple average (Table 6). Only on the 2024 European Parliament election the house effects path won by 0.54 percentage points with an error of 0.99 versus 1.53 for the simple average.

[11] Outer loop refers to the election-level leave-one-out comparison used for final model selection, as opposed to steps like alpha tuning that happen on the remaining elections inside each fold.

Election	Simple poll-only MAE (pp)	House eff. RW poll-only MAE (pp)	Winner	Margin (pp)
2018 parliamentary	0.91	2.34	Simple	1.43
2019 EP	0.97	1.70	Simple	0.73
2022 parliamentary	1.66	3.52	Simple	1.86
2024 EP	1.53	0.99	House eff. RW	0.54
Wins	3	1		

Table 6: Outer LOEO comparison: simple poll-only vs house effects RW poll-only
Source: Compiled by the author.

The simple weighted average won three of the four outer holdouts. However, the house effects path closed the gap substantially on the most recent election, suggesting that as the poll history grows and house effects become better estimated, the richer model may become more competitive. Choosing between the two paths requires balancing empirical and theoretical arguments. On empirical grounds the simple path is clearly stronger. However, this advantage partly reflects data scarcity. When only a few elections are available for a given pollster, the estimated house effect can be unstable, which explains the uneven performance across holdouts. The house effects path came close to matching the simple average on the most recent election, suggesting the richer model becomes more competitive as the poll history grows. On theoretical grounds it is also the better specified model, since it corrects for systematic pollster biases that the simple weighted average cannot separate from true vote share shifts, but with only four elections available, these estimates remain unstable and the model captures noise rather than signal. Given this, the simple weighted average is chosen as the primary forecast, while the house effects path is carried forward as a sensitivity check. The direction of its divergence from the simple combined forecast remains informative even where its magnitude is uncertain.

V.1.5. 2026 Poll Forecast

Table 7 shows the national poll-only forecasts for all evaluated models applied to the 2026 campaign data.

Model	Government (%)	Opposition (%)	Radical opp. (%)	Other (%)
Simple weighted average	40.12	48.68	5.59	5.61
Ridge correction benchmark	50.05	33.96	9.10	6.89
House effects only	41.69	47.05	5.72	5.54
House effects with random walk	39.84	49.38	6.58	4.20

Table 7: 2026 national poll-only forecasts under all evaluated poll models
Source: Compiled by the author.

The simple benchmark placed the opposition at 48.68% and the government at 40.12%, a gap of 8.56 percentage points. The house effects random-walk path widened this slightly to 9.54 points. The Ridge model, which had the worst historical performance, gave an implausibly large government lead and a very high radical opposition share, confirming it is not a reliable tool in this data setting. These four poll-only estimates already show the main sensitivity in the pipeline. The choice of poll model shifts the national picture by several percentage points before the district layer is considered.

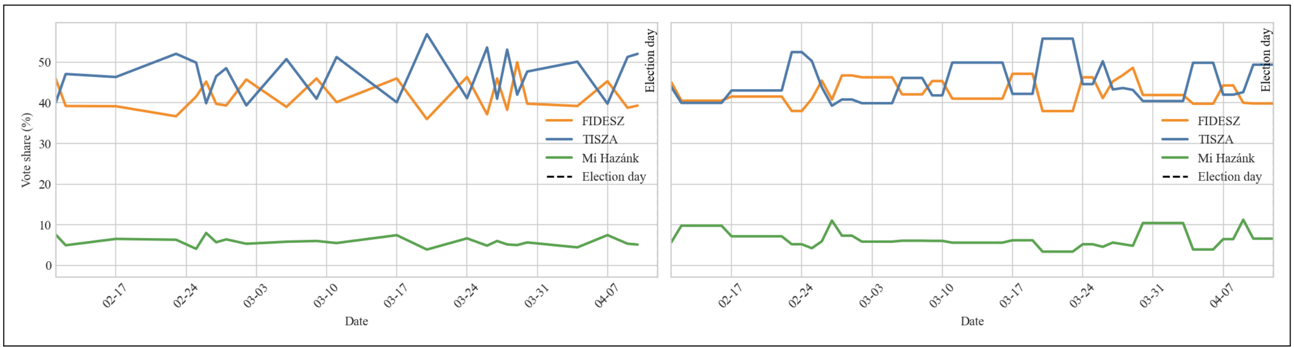

Figure 2: 2026 poll-only vote-share trajectories: simple weighted average (left) and house effects random walk (right)
Source: Compiled by the author.

Figure 2 presents the estimated vote-share trajectories during the 2026 campaign. The left panel shows the simple weighted average. The right panel shows the house-effect-corrected Kalman smoother, which reconstructs a continuous latent support by pooling all available polls after adjusting for pollster-specific biases. In the simple average, the opposition held almost a clear lead throughout the observed window, with government support staying below 50 percent. The Kalman smoother tells a more nuanced story over the full campaign period, with government and opposition starting from closer positions in February 2026 and the gap evolving gradually. Together the two panels illustrate how sensitive raw poll aggregations can be.

V.2. District Fundamentals and Local Lean Results

V.2.1. Prior Structure Selection

The structural staircase compared four nested predictor sets for the national fundamentals prior. The demography-only specification, which used log population, young-age share and old-age share, produced the lowest mean national absolute error at 6.69 percentage points. Adding core economic variables (total unemployment, university-educated unemployment and budget balance) increased the national error to 7.17 points.

Candidate set	Features	Mean national MAE (pp)	Mean district MAE (pp)
Demography only	3	6.69	7.06
+ core economy	6	7.17	7.38
+ modernization	8	8.64	8.80
+ public services, housing	10	8.65	8.80

Table 8: Structural staircase selection results for the national fundamentals prior (Ridge)
Source: Compiled by the author.

The broader modernization and public services layers did not improve fit further and produced the highest errors in the comparison. Because of these, the demography-only set was selected as the prior base. This result is consistent with the role of the prior as a slow-moving structural anchor. The additional economic and infrastructure variables that were expected to be useful in the lean model did not add predictive power here. The likely reason is that the short election history does not provide enough variation for those variables to add reliable signal beyond what the three demographic features already capture.

V.2.2. Prior Form and Alpha Selection

The Ridge and mixed forms were compared on the selected three-feature set. Ridge produced a mean national MAE of 6.69 percentage points and the mixed form 6.78 points. Both forms were nearly identical in practice, but I preferred Ridge because of parsimony (as the methodology specifies). The penalty parameter α was completely insensitive in this setting too. Across the tested range from 2 to 200, the national MAE never varied by more than 0.001 percentage points and the condition number held steady. The α was fixed at 100, providing a conservative degree of shrinkage consistent with the prior layer's structural role, though the actual choice is inconsequential given that the model has little variance to penalize with only four elections and three stable demographic predictors.

V.2.3. 2026 National Fundamentals Prior

After aggregating district-level Ridge predictions to national level, the fundamentals prior gave a considerably more government-friendly picture than the poll model. The prior assigned 50.31% to the government block and 33.45% to the opposition, a gap of 16.9 percentage points. The radical opposition block reached 10.41% and the other block 5.82%.

Source	Government (%)	Opposition (%)	Radical opp. (%)	Other (%)
Fundamentals prior	50.31	33.45	10.41	5.82

Table 9: 2026 national forecast from the fundamentals prior alone (Ridge)
Source: Compiled by the author.

Election	National MAE (pp)
2018 parliamentary	7.11
2019 EP	6.20
2022 parliamentary	7.49
2024 EP	5.98
Mean	6.70

Table 10: Fundamentals prior national MAE across historical holdout elections
Source: Compiled by the author.

The prior consistently overestimated the government block relative to actual results. This is expected because demographics and infrastructure describe long run political geography rather than the current state of a campaign.[12] The 2026 prior continues this pattern, placing the government 5–10 percentage points above what the polls suggest. These two forecasts — the polls and the prior — represent two fundamentally different theories about where the electorate stands. The polls capture recent voter sentiment, while the prior captures the structural map that Hungarian politics has settled into over the past decade. The 70%-30% combination retains both signals and uses the prior mainly as a correction against large short-term poll swings rather than as an independent forecast.

V.2.4. District Lean Model

The penalty parameter α for Ridge was completely insensitive for the lean model as well. All tested values produced identical district MAEs. Contrary to what regularisation would typically achieve, the lean specification appears to leave little to no structure to shrink, likely due to its reduced feature set and the resulting absence of multicollinearity. The lean α was fixed at 30.

The two candidate lean specifications produced clearly different results. The ALR mixed lean model — which uses a county-level random intercept to capture regional political variation — performed better (Table 11).

Model	Mean district MAE (pp)	Mean winner hit rate (%)
Mixed lean model	2.08	91.27
Ridge lean model	2.36	88.92

Table 11: Lean model comparison across LOEO holdouts
Source: Compiled by the author.

The mixed model won on both main criteria, with a gap of 0.28 percentage points in error and 2.35 points in hit rate over the Ridge form. I think these differences are meaningful enough to

[12] In this paragraph, I followed the stylistic and grammatical guidance provided by generative AI.

justify choosing the model with random intercepts. The county-level random intercept captures regional political clustering that a flat Ridge model cannot account for, since districts in the same county tend to share similar political traditions, economic conditions and campaign dynamics.

Based on Table 12, the two middle holdouts (2019 and 2022) are the most stable, with errors below 1.75 points. The slight uptick in 2024 likely reflects the different spatial character of European Parliament elections, where turnout patterns and party identities differ from parliamentary contests. Crucially, there is no upward trend across the four holdouts, which confirms that the model is not over-relying on features that were only informative in the earliest elections.

Election	District MAE (pp)
2018 parliamentary	2.66
2019 EP	1.74
2022 parliamentary	1.73
2024 EP	2.17
Mean	2.08

Table 12: Mixed lean model holdout mean of district MAEs by election

Source: Compiled by the author.

V.2.5. Recent Correction Layer

The recent correction feature set was fixed ex ante — as discussed in Methodology Section IV.2.4. — , including lagged lean variables from the two most recent elections (2022 parliamentary and 2024 European Parliament, three blocks each) and the pro-government digital usage.

On the two available recent elections, the corrected district performance was strong. The winner hit rates were 96.23% and 89.62%, so in each election roughly 100 of the 106 single-member districts guessed correctly the winning blocks.

Election	District MAE after correction (pp)	Winner hit rate (%)
2022 parliamentary	1.99	96.23
2024 EP	2.16	89.62

Table 13: Recent correction layer performance on recent elections (Ridge)

Source: Compiled by the author.

The Ridge penalty α for the recent correction layer was equally insensitive, with all tested values producing identical holdout MAEs, consistent with the pattern observed across the other model layers.

V.2.6. 2026 District Pattern

In the 2026 integrated district forecast, the two 70-30 combined poll paths produce nearly identical spatial outcomes. However, this one district difference between the paths carries more weight than it appears. In a tight race, each additional district win not only adds one seat directly but also reduces the winner's compensation vote pool, shifting list seat allocation in the opposite direction.

Block	Simple combined	House eff. RW combined
Government	56	55
Opposition	50	51
Radical opposition	0	0
Other	0	0

Table 14: Predicted 2026 list-side district winners by forecast

Source: Compiled by the author.

The geographic pattern is consistent across both paths. Opposition support concentrates in Budapest and a small number of urban districts, while the government block maintains a broader advantage across rural and semi-urban areas (Figure 3). This spatial structure is precisely the kind of district heterogeneity the lean model was designed to capture. The house effects random-walk version is shown in Appendix Figure A.5.

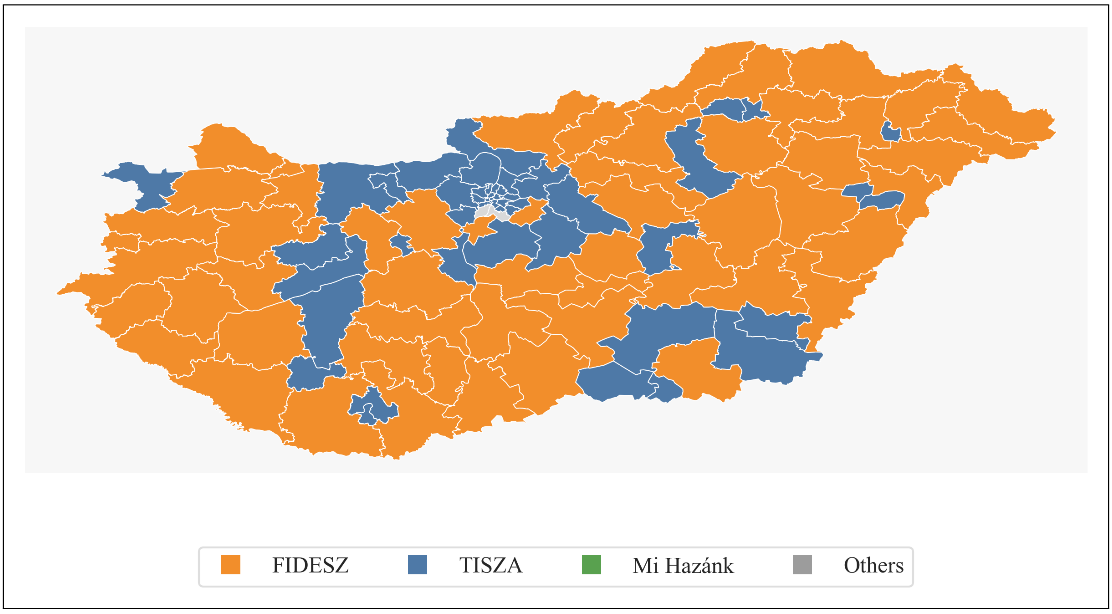

Figure 3: Predicted 2026 district winners under the simple combined forecast

Note: District winners shown here are based on list votes only. The final seat count also uses the separate candidate vote predictions from the candidate-gap model.

Source: Compiled by the author. [13]

V.3. Integrated Forecast, Candidate Effects and Seat Simulation

V.3.1. Poll-Prior Combination

On the simple path, the poll model was already very accurate across all four holdouts. The mean combined error rose to 1.71 percentage points against a poll-only mean of 1.27 points (Table 15). Across the four holdouts, the poll only model therefore has the lower average error.

Election	Poll MAE (pp)	Prior MAE (pp)	Combined MAE (pp)
2018 parliamentary	0.91	7.11	2.75
2019 EP	0.97	6.20	1.29
2022 parliamentary	1.66	7.49	1.52
2024 EP	1.53	5.98	1.28
Mean	1.27	6.70	1.71

Table 15:
Combination backtest MAE by election, simple poll-only vs simple combined
Source: Compiled by the author.

However, the election-level pattern is more informative than the mean. In 2018 and 2019 the polls were really accurate and the prior introduced a large structural error, pulling the combined forecast away from the better signal. But in the two most recent elections the picture reverses. In 2022 the combined forecast (1.52 points) outperformed the poll-only model (1.66 points) and it also performed better in 2024.

Because the prior improved the accuracy in the two most recent cycles — which are the closest cases to 2026 in terms of polling volume, pollster coverage and political context — I will retain the 70-30 combination for the final forecast. Furthermore using the same prior weight in every model variant means that differences in the final forecast come from the poll model itself rather than from changing whether the structural anchor is used. This keeps the sensitivity comparison across variants internally consistent. For transparency, the final results also report the poll only national forecast too.

The house effects path is detailed in the Appendix, see Appendix Table A.6. There the prior also raised the mean combined error, but unlike the simple path there is no consistent improvement in the two most recent elections.

V.3.2. Combined National Forecast

Under the 70% poll and 30% fundamentals combination, the 2026 national forecast for the simple poll path can be seen in Table 16.

[13]Generative artificial intelligence was used for technical assistance in the preparation of this figure and its suggestions were reviewed and verified by the author.

Source	Government (%)	Opposition (%)	Radical opp. (%)	Other (%)
Poll model	40.12	48.68	5.59	5.61
Fundamentals prior	50.31	33.45	10.41	5.82
Combined	43.18	44.11	7.04	5.67

Table 16: 2026 national forecast from the simple poll model, fundamentals prior and combined model

Source: Compiled by the author.

The selected Combined forecast places the opposition at 44.11% and the government at 43.18%, a margin of under one percentage point. This is a substantial compression from the poll-only estimate, where the opposition led by 8.6 points, driven by the fundamentals prior pulling government support upward and opposition support downward. The national race is therefore essentially tied at this stage, with neither main block able to claim a comfortable margin.
At the house effects random-walk path the conclusion is similar (Appendix Table A.7).

V.3.3. National Uncertainty

The simulation needs to draw all three non-base blocks jointly, not independently, because prediction errors across blocks are inevitably related. This correlation structure — which is estimated from the historical ALR forecast errors — can be seen in Table 17. The table shows correlations rather than covariances for easier interpretation.

	Government	Opposition	Radical opposition
Government	1.000	0.396	0.125
Opposition	0.396	1.000	0.136
Radical opposition	0.125	0.136	1.000

Table 17:
Empirical ALR error correlations between blocks from the four-election sample

Note: Positive correlations here reflect the shared ALR baseline, not political alignment between the blocks (see next paragraph).

Source: Compiled by the author.

After applying shrinkage with prior strength 6.0, all three observed correlations are positive and moderate. Since none of the post-shrinkage values exceeds the 0.45 clip threshold, shrinkage alone is the binding regularisation here. When the model misses one block, it tends to miss the others in the same direction. This is expected given the ALR structure. Because all blocks are expressed relative to the other block as the baseline, a systematic miss on that baseline shifts all ALR errors in the same direction at once. For example, if pollsters consistently undercount smaller parties and independents, both the government and the opposition appear inflated in ALR space simultaneously, producing a positive error correlation between them.

V.3.4. Candidate-Gap Model

In Table 18, the separate candidate-gap model outperformed the list proxy overall, despite being trained on only one full parliamentary transition cycle.

This narrow training window required strong regularisation, pulling the predicted candidate gaps substantially toward the list share for all three blocks. But even the shrunken model was still more accurate than the list proxy for both main blocks and this advantage drove the overall result.

Model	Gov. MAE	Opp. MAE	Rad. opp. MAE	Other MAE	Overall MAE
Separate candidate-gap model	1.06	1.29	0.90	1.06	1.08
List proxy	1.32	1.45	0.74	1.39	1.23

Table 18: Grouped county holdout comparison of the single-member-district mean layers
Source: Compiled by the author.

The one exception is the radical opposition, where the list proxy was better. This is consistent with the heavy shrinkage. With only one training cycle and highly volatile Mi Hazánk candidate performance across districts, the model correctly defaults almost entirely to the list share rather than trying to predict a candidate gap it cannot reliably estimate. Therefore the separate model was selected, though for the radical opposition it makes almost no practical difference (so in their case the prediction is nearly the same as the list proxy anyway).

V.3.5. Seat Approximation

The block-level seat approximation was validated on the two available parliamentary elections.

Year	Gov. calc.	Gov. off.	Opp. calc.	Opp. off.	Rad. calc.	Rad. off.	Other calc.	Other off.
2018	127	133	46	37	26	26	0	3
2022	134	135	58	57	7	6	0	1

Table 19: Historical seat reconstruction from the block-level approximation
Note: The Other block is zero here by construction, as explained in Methodology IV.3.4..
Source: Compiled by the author.

The block-level approximation was compared against a rough proxy that simply counts district winners and applies a straight d'Hondt allocation of list seats without the compensation vote mechanism. In Table 20 it can be seen that the mean total absolute seat error was 9 seats for the block-level approximation, while using the rough proxy it is 12 seats. So the block-level approximation was confirmed as the method used in the simulation. The remaining gap between approximated and official seats comes from the block-level modelling approach. Modelling at block rather than at full party-alliance legal detail means that some compensation vote mechanics cannot be replicated exactly. Nevertheless, the approximation is close enough for a meaningful seat distribution.

Seat layer	Mean total absolute error (seats)
Legal-style approximation	9
Rough proxy	12

Table 20: Comparison of the two seat allocation layers on historical elections

Source: Compiled by the author.

V.3.6. Point Forecast and Simulation

The complete forecast results presented in this subsection were archived on Zenodo before election day to establish that the predictions were made before the results were known (Andrási, 2026). There are small decimal and rounding differences between that archived version and values reported here, but they do not affect the substantive forecast or its interpretation.

Table 21 shows the point seat forecast under the main simple weighted average path. At the point estimate, the government block reached a slim majority but fell well short of a two-thirds supermajority, while the opposition did not reach majority. This shows that the race is tight enough for the final parliamentary outcome to be a genuine distributional question.

The other two point seat estimates are shown in Appendix Table A.8.

Government	Opposition	Radical opposition	Other
102	89	8	0

Table 21: 2026 point seat forecast, simple combined forecast

Note: The Other block is zero here by construction, as explained in Methodology IV.3.4..

Source: Compiled by the author.

Table 22 shows the expected seat counts and the 10th–90th percentile ranges from the 1,500-draw Monte Carlo.

Block	Expected seats	P10 seats	P90 seats
Government	94.5	72	116
Opposition	90.6	68	113
Radical opposition	13.8	7	23
Other	–	–	–

Table 22: Expected seats and percentile ranges from the 1,500-draw simulation, simple combined forecast

Source: Compiled by the author.

The P10–P90 range spans roughly 40-45 seats for both main blocks.[14] These wide intervals are a consequence of the Hungarian seat allocation system's sensitivity to small national shifts rather than a model failure. A swing of even a few percentage points can flip marginal districts, each flip reducing the winner's compensation votes and concentrating list seats further, so national shifts compound rather than cancel out. Therefore, point forecasts and simulation means should be read as central estimates within a distribution that still allows a wide range of parliamentary outcomes.

Table 23 shows the scenario shares under the main simple 70% poll and 30% prior path. The corresponding sensitivity scenario shares for the simple poll-only and house effects random walk paths are reported in Appendix Table A.9.

Scenario	Share (%)
FIDESZ finishes first in list share	46.6
TISZA finishes first in list share	53.4
FIDESZ majority (≥100 seats)	39.3
FIDESZ two-thirds majority (≥133 seats)	1.1
TISZA majority (≥100 seats)	29.7
TISZA two-thirds majority (≥133 seats)	1.4

Table 23: Scenario shares from the 1,500-draw seat simulation, simple combined forecast
Source: Compiled by the author.

TISZA holds a slight edge in list share finish probability at 53.4%. However, in Hungary finishing first in list share does not guarantee a parliamentary majority. FIDESZ has a 39.3% chance of a majority and a near-zero chance of a two-thirds supermajority. TISZA reaches majority in 29.7% of draws, despite being more likely to finish first in list share. This gap reflects the district-level structural advantage that FIDESZ retains even when trailing nationally, because the compensation vote system can decouple seat outcomes from raw list share margins. Adding the two majority probabilities, roughly 31% of simulation draws produce a parliament where neither main block holds a majority on its own, making a hung parliament a non-trivial scenario. Neither main block is therefore certain of a majority on its own.

Figure 4 shows the full seat distribution from the simulation under the main simple 70% poll and 30% prior path.

[14] In this paragraph, I followed the stylistic and grammatical guidance provided by generative AI.

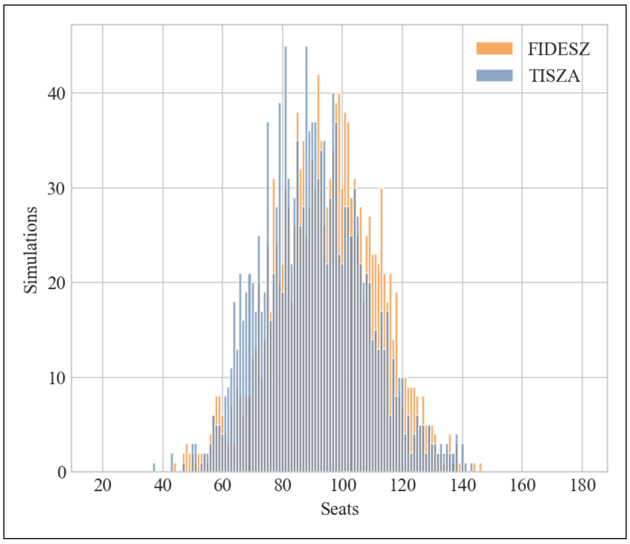

Figure 4: Seat distribution from the 1,500-draw Monte Carlo simulation, simple combined forecast

note: Mi Hazánk is omitted for data visualization clarity

Source: Compiled by the author.

The simple path produces roughly bell-shaped distributions for both main blocks.[15] The FIDESZ distribution peaks closer to the 100-seat majority threshold than the TISZA distribution, whose peak sits slightly further to the left. Both distributions overlap substantially across the 70–120 seat range, so the threshold falls inside the overlap zone rather than cleanly separating the two blocks. This is the visual expression of the asymmetry in Table 23: TISZA is more likely to finish first in list share yet less likely to clear the majority threshold, because its distribution is centered just further from 100 seats.

The corresponding sensitivity results for the poll-only path and the house effects random walk path are shown in Appendix Section VIII.11.. Under the poll-only path TISZA holds a considerably larger seat lead, while the house effects path produces much wider uncertainty intervals, with the P10–P90 range spanning over 100 seats for both main blocks.

V.4. Overall Comparison and Sensitivity Check

This section compares the three forecast variants: the simple poll-only forecast, the simple combined forecast (the primary model throughout the thesis) and the house effects random walk combined forecast.

[15] In this paragraph, I followed the stylistic and grammatical guidance provided by generative AI.

V.4.1. Sensitivity Check

The poll-only forecast isolates the polling signal without the structural prior, while the house effects combined forecast tests how sensitive the conclusions are to the choice of poll model within the combined framework. The three variants span a wide national range: the poll-only path places the opposition ahead by 8.56 percentage points and gives TISZA a 64.3 percent chance of a parliamentary majority, while both combined forecasts compress the gap to under two percentage points and shift the race to an effectively tied outcome, because the structural prior pulls the government share upward by roughly three percentage points regardless of which poll model is used.

The combined paths diverge most sharply on the tail scenarios. Even a marginal national vote share difference between the two paths produces an 18-point gap in the FIDESZ two-thirds supermajority probability (Tables 23 and A.9), because the compensation vote system amplifies small national margins into large seat swings. The poll-only path extends this range further still, reversing the picture entirely so that it is TISZA rather than FIDESZ holding a realistic two-thirds potential. Together the three variants show that the choice of national model does not merely shift the central estimate but determines which block's supermajority is even on the table.

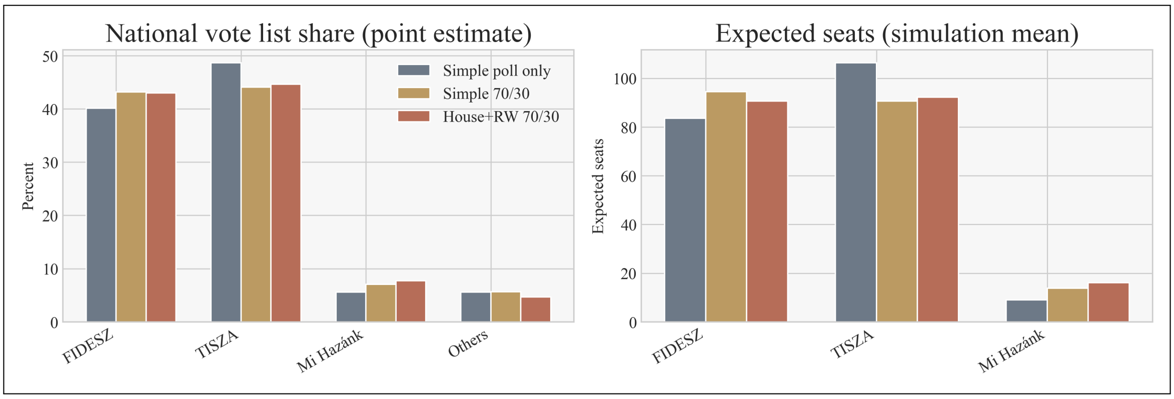

Figure 5: National vote share and expected seats across the three forecast variants
Source: Compiled by the author.

Figure 5 illustrates the prior's role in compressing the race. The left panel shows that the poll-only path gives TISZA a clear national vote share lead, while both combined paths bring the two main blocks to near parity. The right panel shows the same compression in expected seats. The poll-only path gives TISZA a meaningful seat advantage, while under both combined paths FIDESZ and TISZA end up almost level. The two combined paths are nearly indistinguishable in this figure, which is consistent with the small national difference between them, but as the tail scenario analysis shows, that small difference still carries large consequences at the parliamentary threshold.

V.4.2. External Context

External benchmarks are available to contextualise this forecast. The Politico Poll of Polls for Hungary (Politico, 2026) places TISZA at 49 percent and FIDESZ at 39 percent at the time of writing (in early April 2026, shortly before the 12 April 2026 election). This tracker applies a Kalman-filter weighted average of publicly available polls, screening for transparency and representativeness, but it does not appear to publish an explicit pollster-specific house-effects model. The consequence of that design was illustrated in November 2025, when two government-aligned institutes were included in the tracker window, the aggregate briefly showed FIDESZ ahead by one percentage point for the first time in 2025 (Index.hu, 2025). Within days, after those institutes were removed from the window and replaced by a different poll, the apparent lead swung by eight percentage points overnight (Telex, 2025). This episode directly demonstrates the house effects problem that this study tried to address, a raw weighted average is highly sensitive to which institutes happen to publish most recently, regardless of their systematic biases. The Polymarket prediction market (Polymarket, 2026), with more than 58 million US dollars in traded volume at the time of writing, assigned approximately 70 percent probability to TISZA winning the most seats (see Figure A.8 in the Appendix). This is more decisive than the combined forecasts in this thesis, which place the probability of TISZA finishing first in list share at 53 to 54 percent. The divergence most likely reflects that Polymarket incorporates the most recent polls at face value, while this model pulls the national estimate toward the structural prior and corrects for house effects, both of which shift the forecast toward a closer race. The external references point toward a TISZA lead, consistent with the simple poll-only forecast. However, a domestic seat-allocation estimate published by Index (Index.hu, 2026b) similarly projects a close race, consistent with the central range of this model's simulation. Whether the house-effect correction and the structural prior together will produce a more accurate forecast than the raw poll aggregation these external sources primarily rely on is a question only the April 12 result can answer.

V.5. After-election Summary

On the day after the election, the current level of processed votes is 98.94%, at which point the results show a decisive outcome. TISZA won 53% of the national list vote against 38% for the government block, a gap of nearly 15 percentage points. In parliamentary seats the opposition reached 138, above the two-thirds supermajority threshold of 133, while the government fell to 55 seats. The radical opposition obtained 6 seats (National Election Office of Hungary, 2026a).

Path	Gov. error (pp)	Opp. error (pp)	Rad. opp. error (pp)	Other error (pp)	MAE (pp)
Simple poll-only	+1.69	−4.39	−0.24	+2.94	2.32
Simple combined	+4.75	−8.96	+1.21	+3.00	4.48
House eff. RW combined	+4.55	−8.47	+1.90	+2.02	4.24

Table 24: Forecast accuracy: predicted versus actual national vote shares

Note: Signed errors are predicted minus actual. A positive value means the forecast overestimated the block.

Source: Compiled by the author based on National Election Office of Hungary (2026a).

Path	Gov. error	Opp. error	Rad. opp. error	Other error	MAE (seats)
Simple poll-only	+28	−28	0	0	14.0
Simple combined	+47	−49	+2	0	24.5
House eff. RW combined	+42	−45	+3	0	22.5

Table 25: Forecast accuracy: predicted versus actual seat totals

Note: Signed errors are predicted point seats minus actual seats.

Source: Compiled by the author based on National Election Office of Hungary (2026a).

The poll-only and house effects combined forecasts placed TISZA ahead in both vote share and expected seats, while the simple combined predicted a marginal FIDESZ seat advantage, yet all three substantially underestimated the scale of the gap between the two main blocks. The structural prior was the dominant source of error. Trained on demographic patterns from 2018 to 2024, it pulled the combined estimates roughly three percentage points toward FIDESZ support in a year when the government received only 38% of the vote, directly explaining why the poll-only forecast outperformed both combined variants (MAE 2.32 pp vs 4.48 and 4.24 pp). The demographic and economic patterns that defined Hungarian electoral geography over the previous decade did not carry forward to 2026 as much as was expected.

The house effects correction offered a partial improvement, keeping the house effects combined forecast marginally closer to the result than the simple combined. Whether the prior's failure reflects a genuine structural break in Hungarian politics or a prior that was always over-anchored to FIDESZ-era patterns is a question that only future elections can answer. The poll-only's 7.1 percent probability for a TISZA supermajority, against 1.4 percent under the primary combined forecast, is the clearest illustration of how much the prior compressed the forecast away from what the polls were already signalling.

VI. Conclusion

The study set out to answer two questions: whether the outcome of the 2026 Hungarian parliamentary election can be forecast from available data and how sensitive the seat distribution is to small changes in national vote share. The answer to the first question is a qualified yes and the answer to the second is that the seat distribution is extremely sensitive, as the actual result confirmed in the most direct way possible.

The April 12 result showed that the key choice among the forecast variants was whether to include the structural prior at all rather than which poll model to use and the final outcome aligned more closely with the polls. The structural prior was the direct source of the combined variants' additional error, pulling the government estimate upward in a year when the actual result went the other way. Furthermore, the election also set a new voter participation record (HVG, 2026) and the pattern of results suggests that previously inactive voters mobilized disproportionately in favour of TISZA, a shift that any model trained on historical turnout patterns cannot capture in advance.

The seat outcome also answered the second question directly. A large national swing produces an outsized shift in parliamentary seats due to the Hungarian compensation vote system. The model captured this relationship correctly, even though no forecast could have predicted how large the swing would be.

Last but not least, forecasting elections is one of the more demanding exercises in applied statistics, because voters can and do act in ways that no historical record fully anticipates. The 2026 result illustrated this plainly, as even established polling organizations and well-resourced analytical teams largely underestimated the scale of the TISZA wave. The forecast itself does not predict the future. It translates the available evidence into structured probabilities and makes the underlying assumptions explicit and doing that honestly is the most that any forecast can offer.

VII. Bibliography

Aitchison, J. (1986). *The statistical analysis of compositional data*. Chapman & Hall.

Andrási, K. (2026). Pre-election forecast: 2026 hungarian parliamentary election. https://doi.org/10.5281/zenodo.19519258

ATLO Team. (2022a). *Választási térkép 2022*. Átlátszó. Retrieved April 1, 2026, from https://atlo.team/valasztas2022/

ATLO Team. (2022b, January 20). *Így néz ki a kormánypárti hírmédia*. Átlátszó. Retrieved April 1, 2026, from https://atlo.team/igy-nez-ki-a-kormanyparti-hirmedia/

Belsley, D. A., & Klema, V. C. (1974). *Detecting and assessing the problems caused by multicollinearity: A use of the singular-value decomposition* (Working Paper No. 66). National Bureau of Economic Research.

Brown, L. B., & Chappell, H. W. (1999). Forecasting presidential elections using history and polls. *International Journal of Forecasting*, *15*(2), 127–135. https://doi.org/10.1016/S0169-2070(98)00059-4

DKT – Gemius. (2022). *Audience hungary 2022* [Domestic audience aged 16–75, no additional consumer filter].

DKT – Gemius. (2024). *Audience hungary 2024* [Domestic audience aged 16–75, no additional consumer filter].

DKT – Gemius. (2026). *Audience hungary 2026* [Domestic audience aged 16–75, no additional consumer filter].

Erfort, C., Stoetzer, L. F., Gschwend, T., Koch, E., Munzert, S., & Rajski, H. (2026). The zweitstimme forecast for the german federal election 2025: Coalition majorities and vacant districts. *PS: Political Science & Politics*, *59*(1), 48–54. https://doi.org/10.1017/S1049096525000150

Erikson, R. S., & Wlezien, C. (2008). Leading economic indicators, the polls, and the presidential vote. *PS: Political Science & Politics*, *41*(4), 703–707. https://doi.org/10.1017/S1049096508081237

Gelman, A. (2006). Prior distributions for variance parameters in hierarchical models (comment on article by browne and draper). *Bayesian Analysis*, *1*(3), 515–533. https://doi.org/10.1214/06-BA117A

Gelman, A. (2024, August 30). *Why are we making probabilistic election forecasts? (and why don't we put so much effort into them?)* Statistical Modeling, Causal Inference, and Social Science. Retrieved April 8, 2026, from https://statmodeling.stat.columbia.edu/

Gerber, A. S., Gimpel, J. G., Green, D. P., & Shaw, D. R. (2011). How large and long-lasting are the persuasive effects of televised campaign ads? results from a randomized field experiment. *American Political Science Review*, *105*(1), 135–150. https://doi.org/10.1017/S000305541000047X

Government of Hungary. (2025). *Az országgyűlési választások*. Kormányportál. Retrieved April 8, 2026, from https://2010-2014.kormany.hu/hu/mo/a-valasztasi-rendszer/az-orszaggyulesi-valasztasok

Hanretty, C., Lauderdale, B. E., & Vivyan, N. (2018). Comparing strategies for estimating constituency opinion from national survey samples. *Political Science Research and Methods*, *6*(3), 571–591. https://doi.org/10.1017/psrm.2015.79

Heidemanns, M., Gelman, A., & Morris, G. E. (2020). An updated dynamic bayesian forecasting model for the us presidential election. *Harvard Data Science Review*, *2*(4). https://doi.org/10.1162/99608f92.fc62f1e1

Hoerl, A. E., & Kennard, R. W. (1970). Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, *12*(1), 55–67. https://doi.org/10.1080/00401706.1970.10488634

Holland, P. W. (1973). *Weighted ridge regression: Combining ridge and robust regression methods* (Working Paper No. 11). National Bureau of Economic Research.

Hungarian Central Statistical Office. (2026). *Statinfo interactive database*. Retrieved April 5, 2026, from https://statinfo.ksh.hu/Statinfo/themeSelector.jsp?page=2&lang=en

HVG. (2026). *Újabb rekord az urnazárás előtti utolsó részvételi adat*. Retrieved April 13, 2026, from https://hvg.hu/itthon/20260412_ujabb-rekord-az-urnazaras-elotti-utolso-reszveteli-adat-x-szazalek

Index.hu. (2025, November 23). *A politico szerint átvette a vezetést a fidesz a tisza párt előtt, idén először*. Index. Retrieved April 8, 2026, from https://index.hu/belfold/2025/11/23/politico-poll-of-polls-fidesz-tisza-part-mi-hazank-mozgalom-felmeres/

Index.hu. (2026a, March 28). *Visszalépések az ellenzéki oldalon: A tisza párt és a dk-mszp-p párharca*. Index. Retrieved April 1, 2026, from https://index.hu/belfold/2026/03/28/ellenzek-visszalepesek-tisza-part-demokratikus-koalicio-mszp-valasztas2026-orszaggyulesi-valasztasok/

Index.hu. (2026b, April 9). *Választás 2026: Egyéni választókerületek, listás szavazás, mandátumbecslés*. Index. Retrieved April 9, 2026, from https://index.hu/belfold/csatater/2026/04/09/valasztas-2026-egyeni-valasztokeruletek-listas-szavazas-mandatumbecsles-kozter/

Kayser, M. A., Leininger, A., & Vlasenko, A. (2022). A Länder-based forecast of the 2021 german bundestag election. *PS: Political Science & Politics*, *55*(1), 79–84. https://doi.org/10.1017/S1049096521000974

Kiewiet de Jonge, C. P., Langer, G., & Sinozich, S. (2018). Predicting state presidential election results using national tracking polls and multilevel regression with poststratification (mrp). *Public Opinion Quarterly*, *82*(3), 419–446. https://doi.org/10.1093/poq/nfy023

Kóczián, P. (2025, April 21). *Átengedi-e a hatalmát 2026-ban orbán viktor?* Index. Retrieved April 8, 2026, from https://index.hu/velemeny/2025/04/21/valasztas-orban-demokracia-hitelesseg-elszigetelodes/

Kovacs, Z., & Vida, G. (2015). Geography of the new electoral system and changing voting patterns in hungary. *Acta Geobalcanica*, *1*(2), 55–64.

Kovalcsik, T. (2025, December 11). *Mennyit számítanak a levélszavazatok?* Partizán. Retrieved March 31, 2026, from https://www.partizan.hu/post/mennyit-szamitanak-a-levelszavazatok

Lauderdale, B. E., Bailey, D., Blumenau, J., & Rivers, D. (2020). Model-based pre-election polling for national and sub-national outcomes in the us and uk. *International Journal of Forecasting*, *35*(3). https://doi.org/10.1016/j.ijforecast.2019.05.012

Ledoit, O., & Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, *88*(2), 365–411. https://doi.org/10.1016/S0047-259X(03)00096-4

Linzer, D. A. (2013). Dynamic bayesian forecasting of presidential elections in the states. *Journal of the American Statistical Association*, *108*(501), 124–134. https://doi.org/10.1080/01621459.2012.737735

Lock, K., & Gelman, A. (2010). Bayesian combination of state polls and election forecasts. *Political Analysis*, *18*(3), 337–348. https://doi.org/10.1093/pan/mpq002

Martin, G. J., & Yurukoglu, A. (2017). Bias in cable news: Persuasion and polarization. *American Economic Review*, *107*(9), 2565–2599. https://doi.org/10.1257/aer.20160812

National Election Office of Hungary. (2019). *2019 european parliament election results*. Retrieved March 6, 2026, from https://valtor.valasztas.hu/valtort/jsp/orszjkv.jsp?EA=39&W=11

National Election Office of Hungary. (2022). *2022 parliamentary election results*. Retrieved March 31, 2026, from https://vtr.valasztas.hu/ogy2022

National Election Office of Hungary. (2024). *2024 european parliament election results*. Retrieved March 6, 2026, from https://vtr.valasztas.hu/ep2024

National Election Office of Hungary. (2026a). *2026 parliamentary election results*. Retrieved April 13, 2026, from https://vtr.valasztas.hu/ogy2026

National Election Office of Hungary. (2026b). *Általános tájékoztató az országgyűlési képviselők választásáról* [Downloaded PDF in local literature archive on 2026-03-19]. Retrieved April 5, 2026, from https://www.valasztas.hu/altalanos-tajekoztato-az-orszaggyulesi-kepviselok-valasztasarol

National Election Office of Hungary. (2026c). *Electoral statistics downloads*. Retrieved March 6, 2026, from https://vtr.valasztas.hu/stat?tab=letoltesek

National Research Council. (2006). *Asking the right questions about electronic voting*. The National Academies Press. https://www.nationalacademies.org/read/11449/chapter/4

Politico. (2026). *Europe poll of polls: Hungary*. Retrieved April 8, 2026, from https://www.politico.eu/europe-poll-of-polls/hungary/

Polymarket. (2026). *Hungary parliamentary election winner*. Retrieved April 8, 2026, from https://polymarket.com/event/hungary-parliamentary-election-winner

Schäfer, J., & Strimmer, K. (2005). A shrinkage approach to large-scale covariance matrix estimation and implications for functional genomics. *Statistical Applications in Genetics and Molecular Biology*, *4*(1), Article 32. https://doi.org/10.2202/1544-6115.1175

Selb, P., & Munzert, S. (2011). Estimating constituency preferences from sparse survey data using auxiliary geographic information. *Political Analysis*, *19*(4), 455–470. https://doi.org/10.1093/pan/mpr034

Shirani-Mehr, H., Rothschild, D., Goel, S., & Gelman, A. (2018). Disentangling bias and variance in election polls. *Journal of the American Statistical Association*, *113*(522), 607–614. https://doi.org/10.1080/01621459.2018.1448823

Stoetzer, L. F., Neunhoeffer, M., Gschwend, T., Munzert, S., & Sternberg, S. (2019). Forecasting elections in multiparty systems: A bayesian approach combining polls and fundamentals. *Political Analysis*, *27*, 255–262. https://doi.org/10.1017/pan.2018.49

Telex. (2025, November 24). *Fact-check: A politico nem mérte azt, hogy a fidesz vezet a tisza párttal*. Telex. Retrieved April 8, 2026, from https://telex.hu/ellenorzo/2025/11/24/fact-check-politico-brusszel-tisza-fidesz-meres-poll-of-polls

The Economist. (2020, June 11). *How the economist presidential forecast works*. Retrieved March 19, 2026, from https://projects.economist.com/us-2020-forecast/president/how-this-works

Tóka, G. (2018). *The results of the 2018 hungarian parliamentary elections at various levels of aggregation* [Vox Populi blog. Accessed on: 2026-03-06]. Retrieved March 6, 2026, from https://drive.google.com/drive/folders/1o7SRe3YyRWp6LMaIkRiktb16e0oO6f0T?usp=sharing

Tóka, G. (2019a). *The results of the 2019 european parliament elections in hungary at various levels of aggregation* [Vox Populi blog. Accessed on: 2026-03-06]. Retrieved March 6, 2026, from https://drive.google.com/drive/folders/1o7SRe3YyRWp6LMaIkRiktb16e0oO6f0T?usp=sharing

Tóka, G. (2019b). *Vox populi választási kalauz*. Retrieved March 6, 2026, from https://kozvelemeny.wordpress.com/

Tóka, G. (2022). *The results of the 2022 hungarian parliamentary elections at various levels of aggregation* [Vox Populi blog. Accessed on: 2026-03-06]. Retrieved March 6, 2026, from https://drive.google.com/drive/folders/1o7SRe3YyRWp6LMaIkRiktb16e0oO6f0T?usp=sharing

Tóka, G. (2024). *The results of the 2024 european parliament elections in hungary at various levels of aggregation* [Vox Populi blog. Accessed on: 2026-03-06]. Retrieved March 6, 2026, from https://drive.google.com/drive/folders/1o7SRe3YyRWp6LMaIkRiktb16e0oO6f0T?usp=sharing

Wang, W., Rothschild, D., Goel, S., & Gelman, A. (2014). Forecasting elections with non-representative polls. *International Journal of Forecasting*. https://doi.org/10.1016/j.ijforecast.2014.06.001

Weiler, V. (2024, November 19). *Térképen, hogy miként változik meg budapest választókerületi beosztása*. Telex. Retrieved March 31, 2026, from https://telex.hu/belfold/2024/11/19/terkepen-budapest-valasztokeruletei-valtoznak

Wikipedia. (2018). *2018 hungarian parliamentary election*. Retrieved March 31, 2026, from https://en.wikipedia.org/wiki/2018_Hungarian_parliamentary_election

World Bank. (2025). *Consumer price index (2010 = 100) – hungary*. Retrieved April 5, 2026, from https://data.worldbank.org/indicator/FP.CPI.TOTL?locations=HU

YouGov. (2017, May 31). *Yougov election model q&a*. Retrieved April 5, 2026, from https://yougov.com/en-gb/articles/18289-yougov-election-model-q

VIII. Appendix

The analysis was conducted using Python programming environments. The underlying code is available on the study's GitHub repository.

VIII.1. Pollster Activity Summary

This section provides an overview of the polling institutes included in the Vox Populi database. Table A.1 lists the number of unique polls, the latest year of activity and the average sample size for each institute recorded between 2014 and 2026.

Pollster	Unique polls	Latest poll year	Average sample size
Nézőpont	145	2026	1,340.3
Publicus	135	2026	1,033.3
ZRI	100	2026	1,031.2
IDEA	90	2026	1,737.9
Medián	78	2026	1,131.5
Republikon	69	2026	1,026.1
21 Kutató	56	2026	924.1
Unnamed	35	2020	500.0
Századvég	34	2026	2,729.4
Iránytű	28	2026	2,221.8
Real-PR 93	24	2026	965.8
Társadalomkutató	23	2026	1,285.7
e-benchmark	12	2023	954.2
Opinio	11	2026	1,286.4
Alapjogokért Központ	9	2026	1,000.0
Minerva	8	2026	1,039.4
?	6	2026	11,020.0
Századvég/McLaughlin	7	2026	928.6
Pulzus	5	2024	2,400.0
Nézőpont/XXI. Század	5	2026	1,000.0
Medián és 21 Kutató	4	2024	1,000.0
Civitas	3	2021	1,003.7
Tárki	3	2018	1,009.7
Forrás Társadalomkutató Intézet	2	2025	1,000.0
Medián és 21Kutató	2	2023	1,000.0
Psyma	2	2021	1,000.0
MTA TK	2	2020	2,000.0
AtlasIntel	1	2026	1,587.0
ZRI-Republikon	1	2026	1,000.0
ELTE TKK	1	2025	5,000.0
XXI. Század Intézet	1	2025	1,000.0
Ipsos	1	2024	1,025.0
Szondaphone	1	2022	700.0
Uni Wien	1	2022	2,000.0

Table A.1: Pollster activity summary in the Vox Populi polling database

Source: Compiled by the author based on the Vox Populi database.

VIII.2. Descriptive Statistics of Structural Predictors

This section presents the data availability and descriptive statistics for the candidate structural predictors derived from the KSH settlement-level statistics. Table A.2 shows the non-missing share for each election cycle along with the mean, median, standard deviation (SD), minimum (Min) and maximum (Max) values for the pooled sample.

Feature	2018 NA	2019 NA	2022 NA	2024 NA	2026 NA	Mean	Median	SD	Min	Max
Age 0–14 share	100.0%	100.0%	100.0%	100.0%	100.0%	0.145	0.143	0.016	0.089	0.188
Age 65+ share	100.0%	100.0%	100.0%	100.0%	100.0%	0.199	0.198	0.023	0.143	0.258
Blue-collar unemployed per 1,000 residents	100.0%	100.0%	100.0%	100.0%	100.0%	22.386	18.020	16.410	2.071	85.665
Budget balance per 1,000 residents	100.0%	100.0%	100.0%	100.0%	100.0%	5,063.587	2,933.861	65,991.261	-413,058.207	534,617.354
Budget expenditure per 1,000 residents	100.0%	100.0%	100.0%	100.0%	100.0%	344,918.361	301,933.261	249,629.591	112,447.340	3,220,655.113
Budget revenue per 1,000 residents	100.0%	100.0%	100.0%	100.0%	100.0%	349,981.948	316,767.860	249,034.477	115,077.753	3,755,272.467
Cable internet subscriptions per 1,000 residents	84.9%	84.9%	84.9%	84.9%	84.9%	130.517	132.287	49.102	12.828	301.992
Cars per 1,000 residents	100.0%	100.0%	100.0%	100.0%	100.0%	395.535	394.152	64.516	224.710	676.379
Consumer price index	84.9%	84.9%	84.9%	84.9%	84.9%	130.861	121.642	24.476	111.384	177.341
Doctors per 1,000 residents	100.0%	100.0%	100.0%	100.0%	100.0%	0.609	0.605	0.086	0.387	0.851
Dwellings built per 1,000 residents	100.0%	100.0%	100.0%	100.0%	99.1%	1.732	1.011	2.002	0.011	13.361
Dwellings terminated per 1,000 residents	99.1%	97.2%	90.6%	94.3%	96.2%	0.206	0.161	0.181	0.000	1.046
General practitioners per 1,000 residents	100.0%	100.0%	100.0%	100.0%	100.0%	0.469	0.471	0.072	0.308	0.675
Internet subscriptions per 1,000 residents	84.9%	84.9%	84.9%	84.9%	84.9%	309.065	312.018	58.835	154.882	475.407
Log population	100.0%	100.0%	100.0%	100.0%	100.0%	11.416	11.417	0.114	11.107	11.683
Municipal dwellings built per 1,000 residents	25.5%	21.7%	32.1%	32.1%	26.4%	0.046	0.010	0.114	0.000	0.984
New car registrations per 1,000 residents	100.0%	100.0%	100.0%	100.0%	100.0%	31.351	29.466	12.636	12.117	105.306
Nominal labour income per 1,000 residents	84.9%	84.9%	84.9%	84.9%	84.9%	9,754.472	7,558.839	6,545.980	1,554.562	36,451.923
Optical internet subscriptions per 1,000 residents	82.1%	84.0%	84.9%	84.9%	84.9%	96.523	93.555	67.071	0.013	260.060
Other internet subscriptions per 1,000 residents	84.9%	84.9%	84.9%	84.9%	84.9%	19.510	17.071	13.333	0.419	78.251
Pediatric doctors per 1,000 residents	100.0%	100.0%	100.0%	100.0%	100.0%	0.140	0.138	0.041	0.031	0.246
Permanent age 0–24 share	100.0%	100.0%	100.0%	100.0%	100.0%	0.255	0.252	0.030	0.175	0.337
Permanent female age 0–14 share	100.0%	100.0%	100.0%	100.0%	100.0%	0.072	0.070	0.009	0.046	0.099
Permanent female age 65+ share	100.0%	100.0%	100.0%	100.0%	100.0%	0.122	0.122	0.015	0.089	0.160
Permanent male age 0–14 share	100.0%	100.0%	100.0%	100.0%	100.0%	0.076	0.075	0.009	0.049	0.102
Permanent male age 65+ share	100.0%	100.0%	100.0%	100.0%	100.0%	0.077	0.076	0.010	0.051	0.118
Permanent population share	100.0%	100.0%	100.0%	100.0%	100.0%	1.026	1.031	0.035	0.920	1.133
Post offices per 1,000 residents	100.0%	100.0%	100.0%	100.0%	100.0%	0.262	0.241	0.154	0.026	0.690
Private dwellings built per 1,000 residents	100.0%	100.0%	100.0%	99.1%	99.1%	0.756	0.468	0.885	0.000	6.762
Real labour income per 1,000 residents	84.9%	84.9%	84.9%	84.9%	84.9%	7,335.514	5,924.340	4,370.102	1,395.673	20,554.693
TV subscriptions per 1,000 residents	0.0%	0.0%	84.9%	84.9%	84.9%	358.799	358.214	37.259	212.994	460.938
Unemployed 180+ days per 1,000 residents	100.0%	100.0%	100.0%	100.0%	100.0%	13.619	10.703	9.719	1.070	49.965
Unemployed 365+ days per 1,000 residents	100.0%	100.0%	100.0%	100.0%	100.0%	8.329	6.503	5.978	0.698	33.861
Unemployed people per 1,000 residents	100.0%	100.0%	100.0%	100.0%	100.0%	26.924	22.318	16.866	5.266	91.827
University-educated unemployed per 1,000 residents	100.0%	100.0%	100.0%	100.0%	100.0%	0.762	0.608	0.524	0.125	2.882
Young unemployed per 1,000 residents	100.0%	100.0%	100.0%	100.0%	100.0%	2.152	1.318	2.239	0.030	11.787
xDSL internet subscriptions per 1,000 residents	84.9%	84.9%	84.9%	84.9%	84.9%	63.373	60.503	31.132	7.528	153.012

Table A.2: Non-missing share and pooled descriptive statistics of structural variables

Source: Compiled by the author based on KSH data.

VIII.3. Media Consumption Data Availability

This section summarizes the data availability for the online media outlets in its original form. To keep the table manageable and to avoid an overly complex table, the metrics (real users, page views and average time per view) and the time windows are aggregated by outlet. Table A.3 shows the minimum non-missing percentage across these metrics for each outlet and election year. This conservative approach ensures that any partial data gaps within an outlet or time window are clearly visible.

Media outlet	2022	2024	2026
24	100.0%	100.0%	100.0%
444	100.0%	100.0%	100.0%
Blikk	100.0%	100.0%	100.0%
Borsonline	100.0%	100.0%	100.0%
Facebook	0.0%	100.0%	100.0%
Hirado	100.0%	100.0%	100.0%
Hvg	100.0%	100.0%	100.0%
Index	100.0%	100.0%	100.0%
Infostart	100.0%	100.0%	100.0%
Instagram	0.0%	100.0%	100.0%
Magyarnemzet	100.0%	100.0%	100.0%
Mandiner	100.0%	100.0%	100.0%
Mediaklikk	100.0%	100.0%	100.0%
Nepszava	100.0%	100.0%	100.0%
Origo	100.0%	100.0%	100.0%
Ripost	100.0%	100.0%	100.0%
Rtl	100.0%	100.0%	100.0%
Szerencsejatek	0.0%	100.0%	100.0%
Szeretlekmagyarorszag	100.0%	100.0%	100.0%
Telex	100.0%	100.0%	100.0%
Tenyek	100.0%	100.0%	100.0%
Tiktok	0.0%	100.0%	100.0%
Tippmixpro	0.0%	0.0%	100.0%
Tv2	100.0%	100.0%	100.0%

Table A.3: Media availability minimum non-missing percentage by year (aggregated)

Source: Compiled by the author based on Gemius audience data.

VIII.4. Polling Trends of Smaller Parties in 2026

This section details the polling results for the smaller parties that were excluded from the primary opposition block in the 2026 forecast. As discussed in Section III.2.3., these parties (DK–MSZP–P and MKKP) have consistently polled below the 5% parliamentary threshold in the most recent surveys. Table A.4 summarizes the support levels across various institutes for the total population, while Figure A.1 provides a visual comparison of these trends.

Poll (Pollster and Date)	DK-MSZP-P	MKKP
21 Kutató (03-06)	1.0	1.0
21 Kutató (03-06)	1.0	1.0
Minerva (03-11)	1.1	1.4
Nézőpont (03-17)	3.0	3.0
Medián (03-20)	1.0	2.0
Medián (03-20)	1.0	2.0
Nézőpont (03-24)	3.0	3.0
Nézőpont (03-24)	3.5	3.5
Republikon (03-26)	2.0	3.0
Republikon (03-26)	2.0	4.0
21 Kutató (03-28)	1.0	1.0
21 Kutató (03-28)	1.0	1.0
ZRI (03-28)	3.0	4.0
ZRI (03-28)	3.0	3.0
Publicus (03-30)	3.0	3.0
IDEA (04-04)	3.0	2.0
Iránytű (04-04)	1.0	4.0
Iránytű (04-04)	1.0	3.0
Publicus (04-09)	2.0	2.0
AtlasIntel (04-10)	1.5	1.4

Table A.4: Support for smaller parties in the latest 20 polls of 2026 (total population)

Source: Compiled by the author based on the latest 2026 polls.

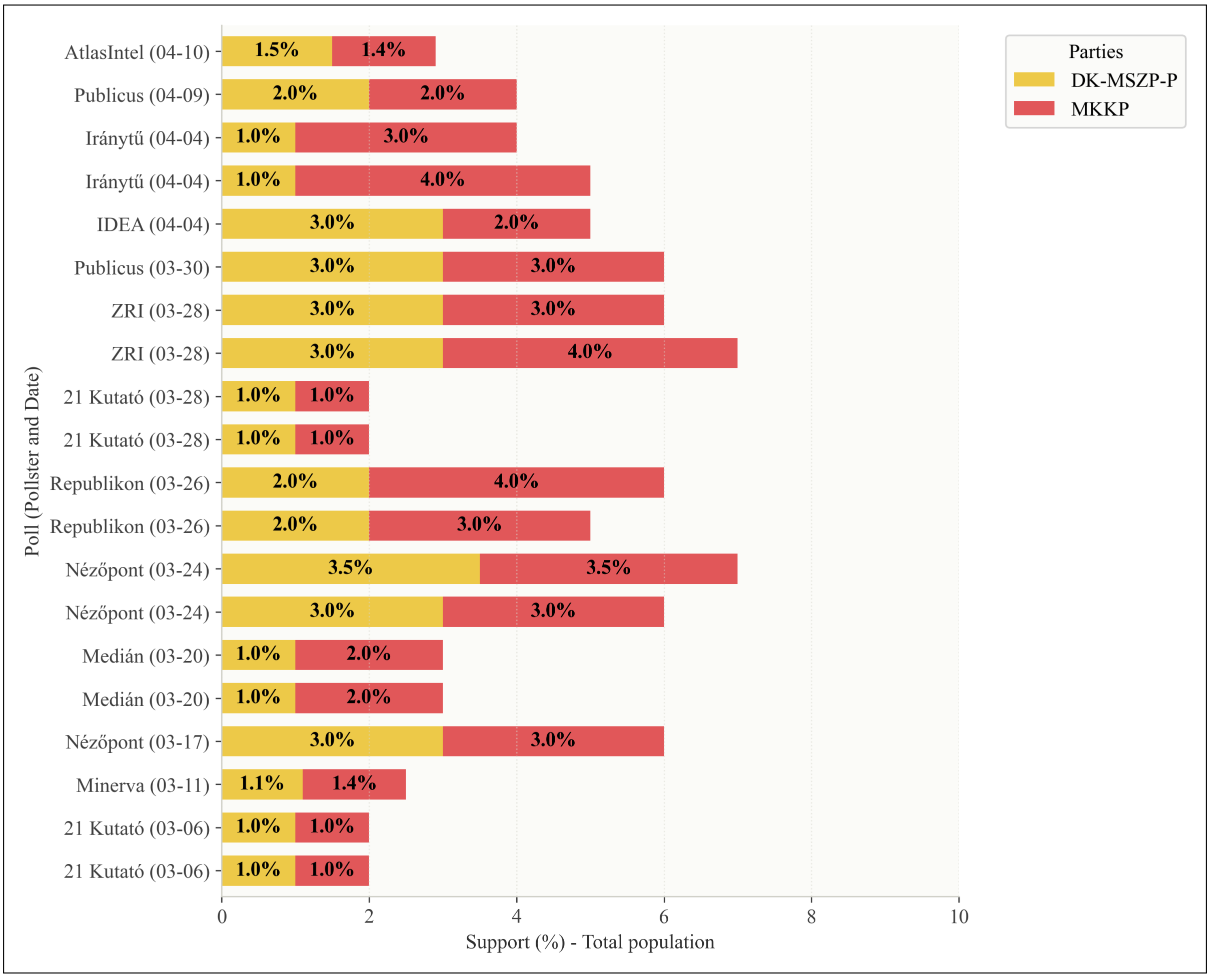

Figure A.1: Pollster comparison of smaller parties support in 2026 (total population)

Source: Compiled by the author based on the latest 2026 polls.

VIII.5. Filtered Pollster Activity Summary

This section summarizes the polling activity during the campaign period. Table A.5 includes only those polls that were conducted within 365 days of an election and had full information on sample size, interview mode and respondent base.

Pollster	Unique polls	Latest poll year	Average sample size
Republikon	29	2026	1,017.2
Publicus	28	2026	1,003.4
ZRI	28	2026	1,053.6
IDEA	27	2026	1,666.7
Nézőpont	25	2026	1,470.0
Medián	19	2026	1,147.4
Iránytű	14	2026	1,358.5
21 Kutató	11	2026	1,163.6
e-benchmark	10	2022	1,000.0
Századvég	9	2024	1,000.0
Alapjogokért Központ	7	2026	1,000.0
Minerva	7	2026	1,045.0
Társadalomkutató	7	2026	1,428.6
Real-PR 93	6	2025	1,000.0
Unnamed	6	2018	500.0
Nézőpont/XXI. Század	5	2026	1,000.0
Századvég/McLaughlin	5	2026	1,000.0
Forrás Társadalomkutató Intézet	2	2025	1,000.0
Civitas	2	2021	1,002.0
Psyma	2	2021	1,000.0
Tárki	2	2018	1,014.5
AtlasIntel	1	2026	1,587.0
ELTE TKK	1	2025	5,000.0
Ipsos	1	2024	1,025.0

Table A.5: Pollster activity summary in the campaign-period polling database (filtered)
Source: Compiled by the author based on the Vox Populi database.

VIII.6. Correlation Matrix of Structural Predictors

This section provides the correlation matrix for the ten structural variables selected for the final model. The matrix shows the Pearson correlation coefficients, indicating the strength and direction of linear relationships between the predictors.

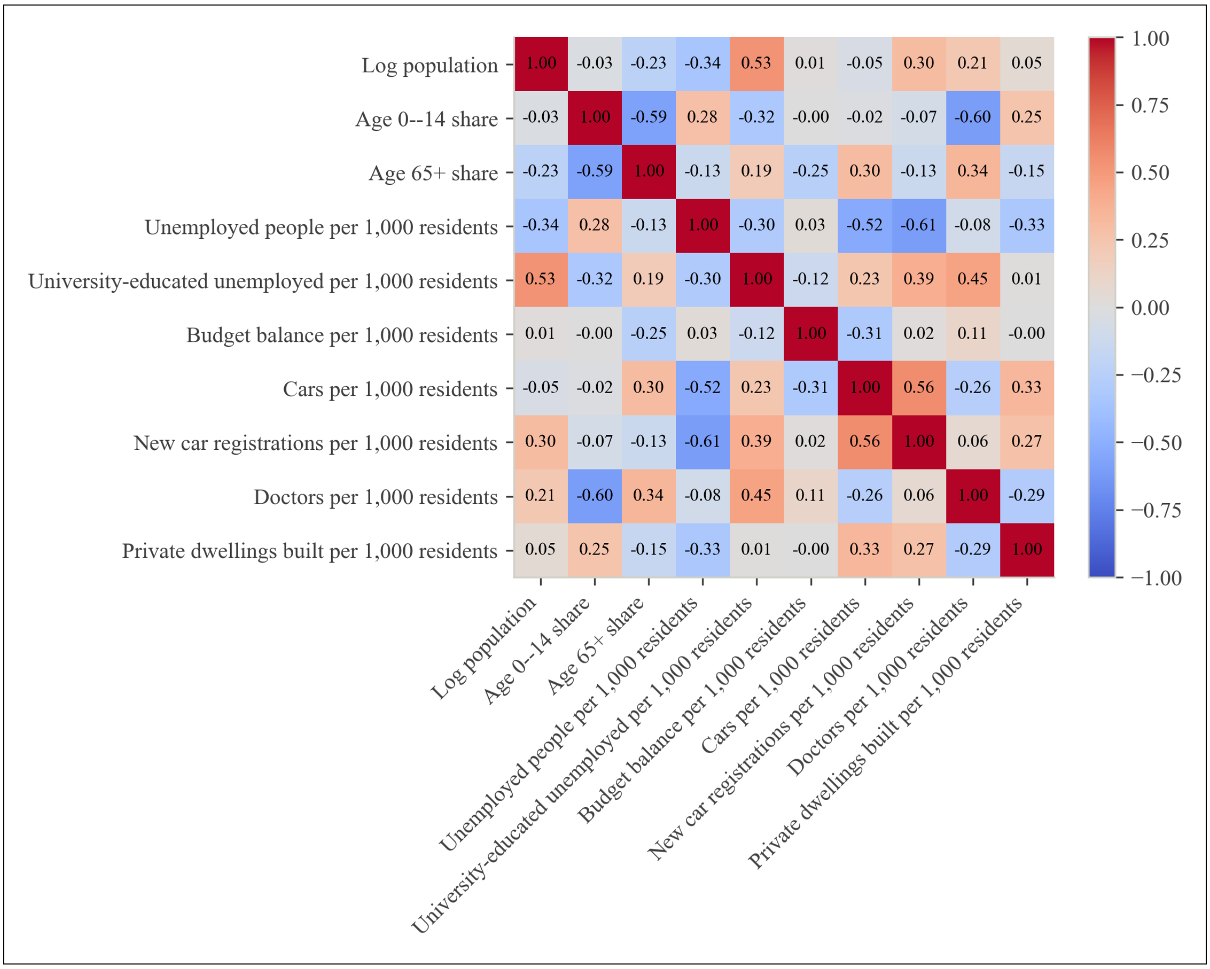

Figure A.2: Correlation matrix of the structural predictor variables

Note: The matrix displays Pearson correlation coefficients. Higher absolute values represent stronger linear dependencies among the variables.
Source: Compiled by the author based on KSH settlement-level data.

VIII.7. Distribution of Media Predictors

This section visualizes the effect of the per-capita log transformation on the media variables. Figure A.3 compares the distribution of the raw ratio (raw user metric divided by total population) with the final log-transformed predictor. The transformation effectively reduces the heavy right-skewness of the raw audience data, resulting in a more balanced distribution for modelling.

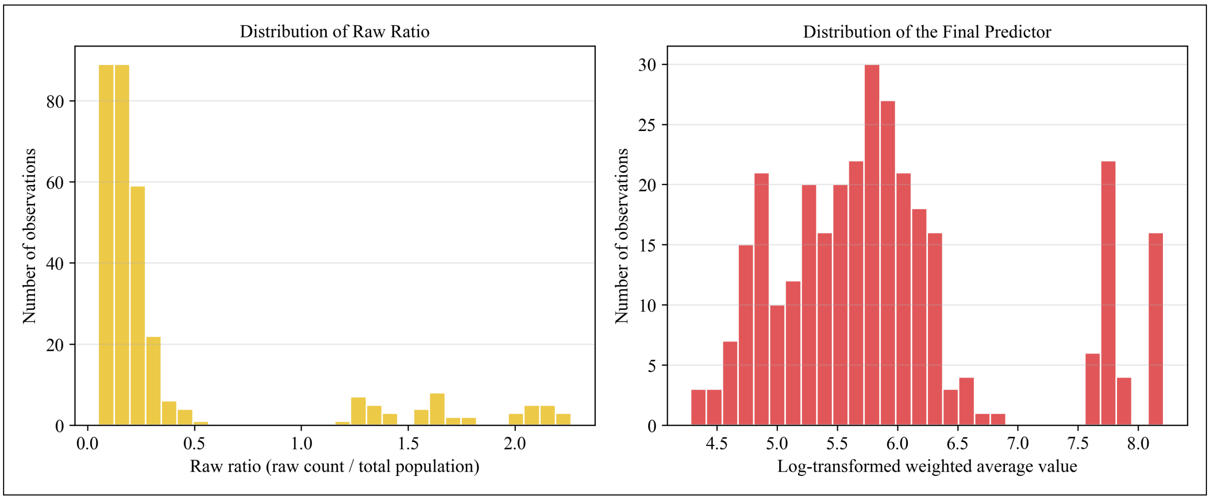

Figure A.3: Distribution of raw vs. final media predictor

Source: Compiled by the author based on Gemius audience data.

VIII.8. Media Signal Strength Across Metrics and Windows

Figure A.4 displays the mean absolute correlation between various media metrics (real users, page views and average time per view) and the government's vote margin. The analysis confirms that *real_users* consistently provides the strongest signal and that recent time windows maintain their reliability compared to earlier periods.

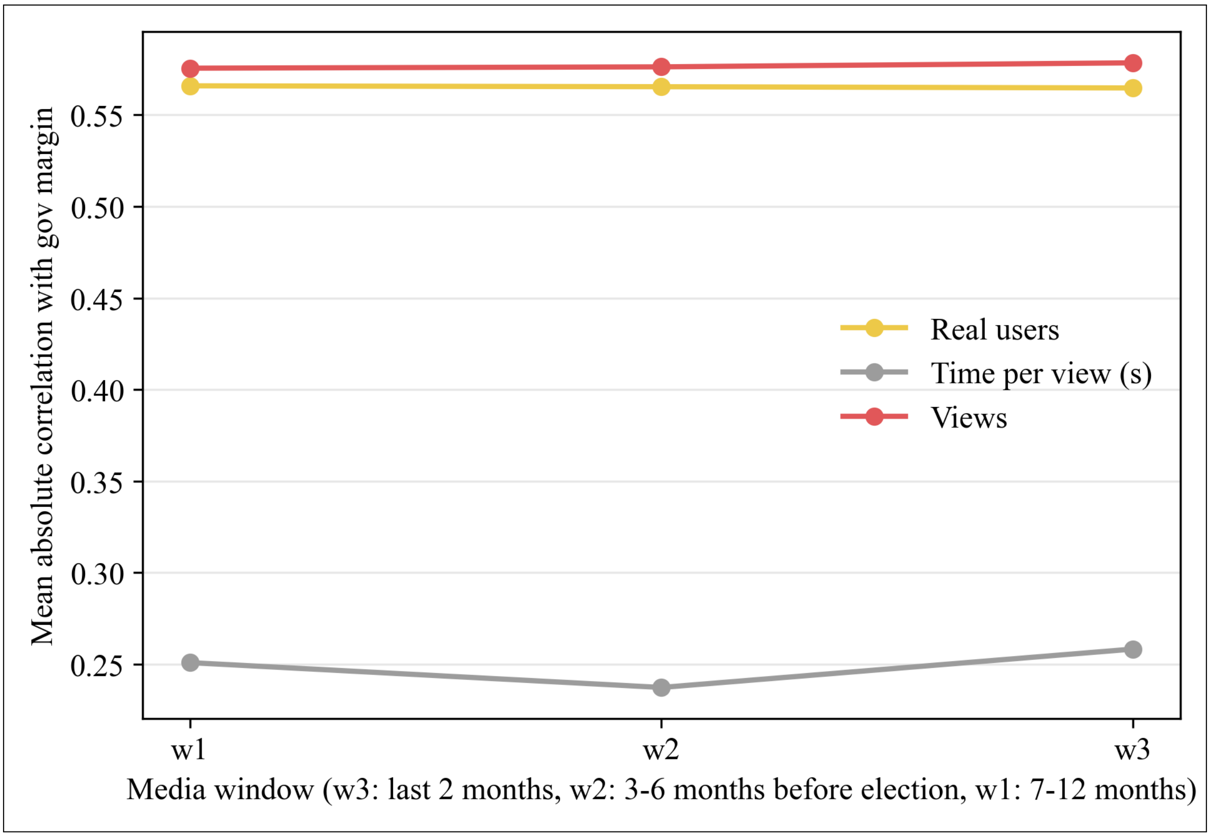

Figure A.4: Mean absolute correlation with government margin by media metric and window

Source: Compiled by the author based on Gemius audience and election data.

VIII.9. House Effects Random Walk Combined Forecast: District Winner Map

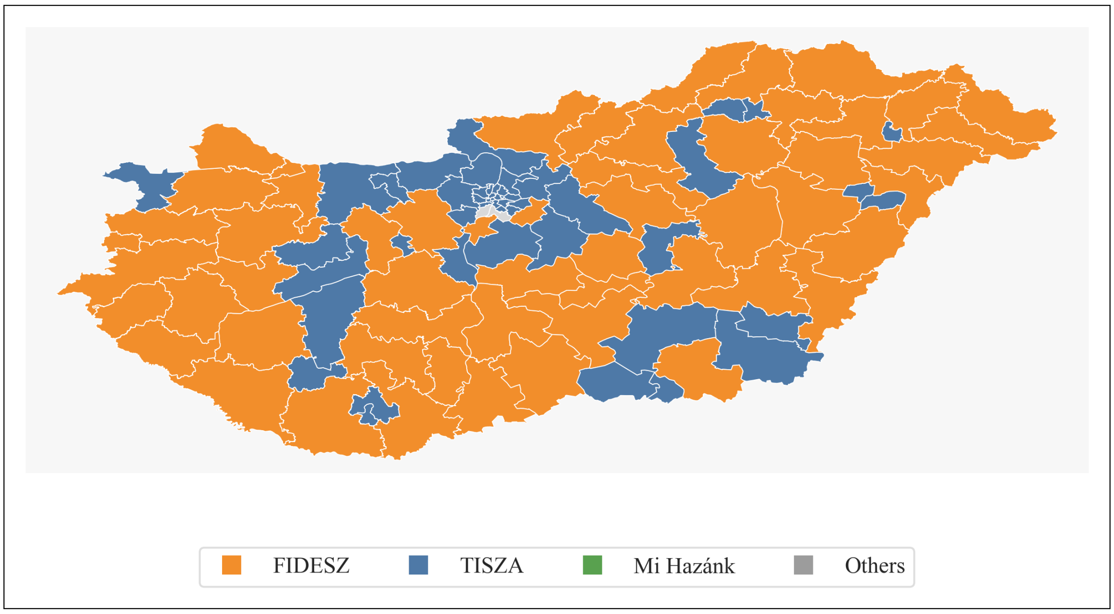

Figure A.5: Predicted 2026 district winners under the house effects random walk combined forecast

Note: District winners shown here are based on list votes only. The final seat count also uses the separate candidate vote predictions from the candidate-gap model.

Source: Compiled by the author. [16]

VIII.10. Sensitivity National Forecast Table

Table A.7 reports the 2026 national forecast for the house effects random walk sensitivity path.

Election	Poll MAE (pp)	Prior MAE (pp)	Combined MAE (pp)
2018 parliamentary	2.34	7.11	2.77
2019 EP	1.70	6.20	1.10
2022 parliamentary	3.52	7.49	2.95
2024 EP	0.99	5.98	2.49
Mean	2.14	6.69	2.33

Table A.6: Combination backtest MAE by election, house effects RW poll-only vs house effects RW combined

Source: Compiled by the author.

[16] Generative artificial intelligence was used for technical assistance in the preparation of this figure and its suggestions were reviewed and verified by the author.

Source	Government (%)	Opposition (%)	Radical opp. (%)	Other (%)
Poll model	39.84	49.38	6.58	4.20
Fundamentals prior	50.31	33.45	10.41	5.82
Combined	42.98	44.60	7.73	4.69

Table A.7: 2026 national forecast: house effects RW poll-only, fundamentals prior and house effects RW combined

Source: Compiled by the author.

VIII.11. Sensitivity Seat Distributions

Table A.8 reports the alternative point seat forecasts for the simple poll-only and house effects random walk paths. Table A.9 reports the corresponding scenario shares for these two sensitivity paths.

Path	Government	Opposition	Radical opposition	Other
Simple poll-only	83	110	6	0
House eff. RW combined	97	93	9	0

Table A.8: 2026 point seat forecasts under the simple poll-only and house effects RW combined forecasts

Note: The Other block is zero here by construction, as explained in Methodology IV.3.4..

Source: Compiled by the author.

Scenario	Simple poll-only (%)	House eff. RW combined (%)
FIDESZ finishes first in list share	18.9	46.6
TISZA finishes first in list share	81.1	53.3
FIDESZ majority (≥100 seats)	18.7	42.4
FIDESZ two-thirds majority (≥133 seats)	0.1	19.1
TISZA majority (≥100 seats)	64.3	42.9
TISZA two-thirds majority (≥133 seats)	7.1	19.4

Table A.9: Scenario shares from the 1,500-draw seat simulation, simple poll-only and house effects RW combined

Source: Compiled by the author based on the final 2026 forecast export.

Block	Expected seats	P10 seats	P90 seats
Government	83.5	62	106
Opposition	106.4	83	129
Radical opposition	9.1	2	16
Other	–	–	–

Table A.10: Expected seats and percentile ranges from the 1,500-draw simulation, simple poll-only

Source: Compiled by the author.

Block	Expected seats	P10 seats	P90 seats
Government	90.7	37	146
Opposition	92.2	38	148
Radical opposition	16.2	5	31
Other	–	–	–

Table A.11: Expected seats and percentile ranges from the 1,500-draw simulation, house effects RW combined

Source: Compiled by the author.

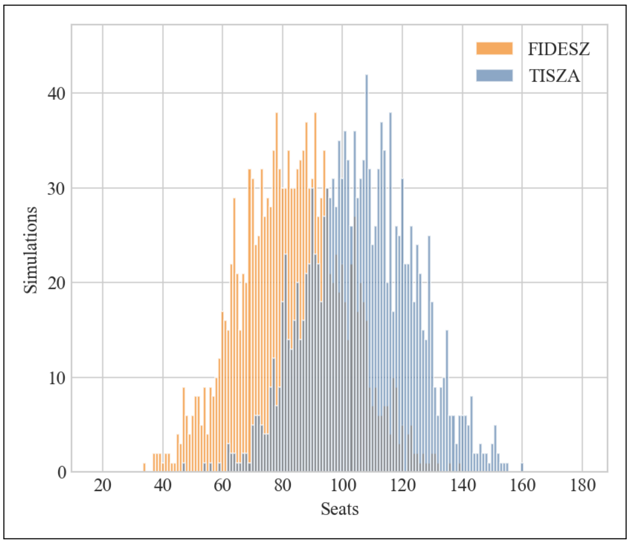

Figure A.6: Seat distribution from the 1,500-draw Monte Carlo simulation, simple poll-only

note: Mi Hazánk is omitted for data visualization clarity

Source: Compiled by the author.

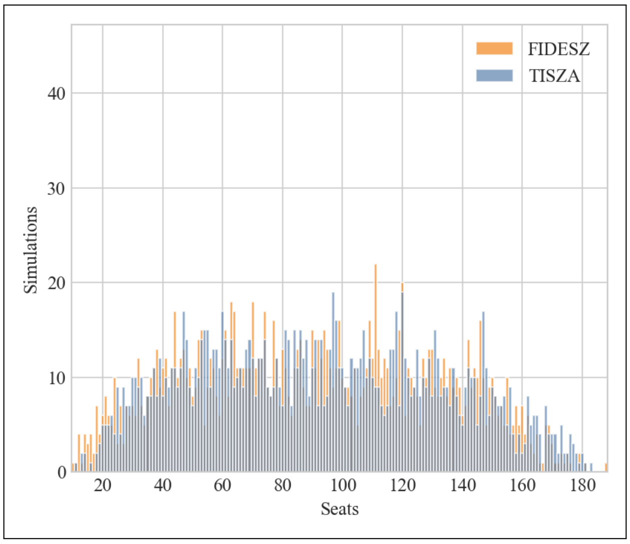

Figure A.7: Seat distribution from the 1,500-draw Monte Carlo simulation, house effects RW combined

note: Mi Hazánk is omitted for data visualization clarity
Source: Compiled by the author.

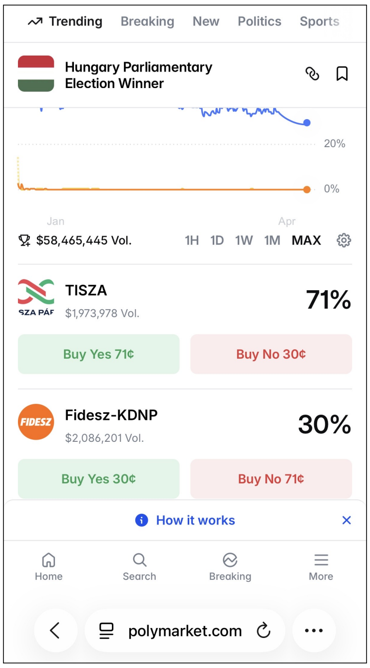

Figure A.8: Polymarket prediction market for the 2026 Hungarian parliamentary election, screenshot taken on 8 April 2026
Source: Polymarket (Polymarket, 2026).

VIII.12. Summary of Model Selection Decisions

Table A.12 collects all model and parameter selection decisions made across the forecast.

Forecast stage	Selected specification	Selection basis
National poll model (main)	Simple weighted average	LOEO: 3 out of 4 outer holdout wins
National poll model (sensitivity)	House effects with random walk	Close 2024 result, theoretical value
Poll α	20	Condition number, flat MAE sensitivity
Prior structure	Demography only (3 features)	LOEO national MAE: 6.69 pp
Prior form	Ridge	Marginal advantage over mixed (6.69 vs 6.78 pp)
Prior α	100	Flat sensitivity, conservative shrinkage
Lean model form	ALR mixed lean model	Clear holdout advantage: 2.08 vs 2.36 pp
Lean α	30	Flat sensitivity
Recent correction	Lag plus media, $\lambda = 1.0$	Fixed ex ante (only 2 recent elections)
Poll-prior combination	Fixed 70% poll weight	Literature benchmark, too few elections to tune
Candidate-gap layer	Separate candidate-gap model	Grouped holdout: 1.08 vs 1.23 pp for list proxy
Seat approximation	Block-level rule approximation	Historical error: 9 vs 12 seats for rough proxy

Table A.12: Summary of all model selection decisions in the forecast

Source: Compiled by the author.

IX. Declaration on the Use of Artificial Intelligence

I, Andrási Kristóf (Neptun code: CF4NE3), declare, in awareness of my ethical responsibility under the Code of Ethics of Corvinus University of Budapest, that I used generative artificial intelligence systems and services in the preparation of this thesis as follows: ChatGPT GPT-5.4, operated by OpenAI (https://chatgpt.com) and Claude Sonnet 4.6, operated by Anthropic (https://claude.ai).

Specifically, I used these tools for the following purposes:

- debugging programming code
- searching within longer documents, with every result checked manually
- translating selected parts of more complex foreign language works into my native language, to verify that I had understood the original meaning correctly
- improving the style and grammar of the thesis text

Following the use of these tools, all generated content was thoroughly reviewed and edited by me. I assume full responsibility for the final contents of this manuscript.